Article

Representation-layer leakage inflates few-shot and cross-domain evaluation under distribution shift

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Evaluating few-shot and cross-domain generalisation under distribution shift is a ubiquitous step in the modern artificial-intelligence paradigm of "first learn a representation, then report performance on held-out classes/domains". We characterise a subtle yet systematic flaw within it—**representation-layer evaluation leakage**: when the representation encoder is learned on data that include the held-out evaluation domain, the measured few-shot accuracy is systematically inflated. We formalise it under a linear-Gaussian nearest-prototype model, **prove that this leakage bias is non-negative**, and give a zero-cost, principled remedy—making the training set for representation learning strictly disjoint from the held-out evaluation domain (the **fold-restricted protocol**). On an in-house nine-domain apple near-infrared benchmark together with five additional public spectral/hyperspectral datasets (spanning five types of shift: instrument, season, year, country and space), a mixed-effects model with dataset as a random effect yields a robustly positive overall leakage ($\Delta = 0.132$, positive in 6/6 datasets), indicating that this is a phenomenon general across modalities and domains. A four-encoder decomposition further delimits it honestly: the leakage is **driven mainly by label-aware exposure** ($\Delta_{\text{sup}} = +0.084$), whereas the purely unlabelled exposure component is small ($\Delta_{\text{unsup}} = +0.012$, confidence interval spanning zero); leakage also changes which method is selected as best on nearly half of the evaluation instances. Evaluation leakage is a general risk across spectroscopy, remote sensing, medical imaging and genomics; we argue that the difference between the fold-restricted and the shared protocol should be reported as a standard item of few-shot and cross-domain evaluation—a low-cost, broadly adoptable upgrade in evaluation hygiene.

1 Introduction

The apple (*Malus domestica* Borkh.) is one of the most widely consumed temperate fruits in the world, and Chinese production accounts for more than 47% of the global supply (FAO, 2022). Behind this figure lie considerable differences in the economic value attached to origin: the price gap between Xinjiang Fuji and Shandong Red Fuji is sometimes widened markedly by nothing more than a certificate of origin. Geographical-indication pricing, food-safety traceability and the fight against false labelling appear to have different emphases, yet all point to the same question: how can the origin of an apple be reliably determined? Conventional means—stable-isotope-ratio analysis, multi-element fingerprinting and sensory evaluation—each rest on a mature and rigorous analytical framework, but they depend on specialised laboratories and take hours per determination, which makes it difficult to meet the turnaround requirements of batch-scale customs screening (Danezis, Tsagkaris, Camin, Brusic and Georgiou, 2016).

Against this background, near-infrared (NIR) diffuse-reflectance spectroscopy is uniquely attractive: it requires no consumables, and a single scan of less than a minute yields a directly interpretable spectrum (Nicolai, Beullens, Bobelyn, Peirs, Saeys, Theron and Lammertyn, 2007; Walsh, Blasco, Zude-Sasse and Sun, 2020). The sources of its spectral variation are manifold—soluble solids, organic acids, polyphenols and water content jointly reflect the geochemical background, climatic rhythm and agronomic practice of the production region, forming an origin-related "fingerprint" with a certain discriminative power in the high-dimensional spectral space (Cen and He, 2007; Woodcock, Downey, Kelly and O'Donnell, 2007). It should be noted that, for commercial apples from wholesale markets, NIR spectra simultaneously encode combined information on variety, storage conditions, ripeness and supply-chain handling; the "origin traceability" referred to in this paper is therefore, strictly speaking, "origin-related discrimination under commercial-batch conditions" rather than direct detection of a purely geochemical compositional signal. Partial least squares discriminant analysis (PLS-DA) is the classical tool for such tasks and usually gives satisfactory accuracy under controlled single-year conditions (Geladi and Kowalski, 1986); as data become more plentiful, support vector machines (SVMs) and convolutional neural networks have begun to surpass PLS-DA in some settings, continually pushing out this performance frontier (Cortes and Vapnik, 1995; Ekramirad, Khaled, Doyle, Loeb, Donohue, Villanueva and Adedeji,

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2022). In recent years, NIR/FTIR spectroscopy combined with chemometrics and deep learning has been widely used for origin traceability and authenticity identification of agricultural products such as tea, edible oil and soybean (Liu, Huang, Li, Chen, Cui, Lu, Wang, Li, Xu, Zhong and Ning, 2022; Ye and Meng, 2022; Han, Wang, Xu, Wang, Dou and Lan, 2026); for the methodological evolution see a recent review (Walsh, Neupane, Koirala, Li and Anderson, 2023).

There is, however, a key obstacle here: models that perform well in single-year controlled experiments are often difficult to transfer to a new harvest season. NIR models trained on controlled laboratory data routinely exceed 90% accuracy on same-year test sets, but when deployed in the following year, climate, soil nutrient status and instrument response all drift, and the gap between the training distribution and the deployment distribution becomes apparent (Feudale, Woody, Tan, Myles, Brown and Ferre, 2002; Li, Huang, Wang, Liu, He and Fan, 2022a; Pan, Li, Hu, Zhang and Zhao, 2024a). Dedicated studies of such "cross-year generalisation" in the apple NIR field are in fact rather limited: most studies remain confined to a single year, and cross-year performance is at most mentioned in passing as a secondary result. Existing calibration-transfer methods—direct standardisation (DS), piecewise direct standardisation (PDS) and gradient-boosted recalibration alike—were originally designed for instrument-to-instrument transfer, and most of them require a batch of labelled target samples to be prepared on the deployment side, a prerequisite that is often hard to meet when a new season arrives (Fearn, 2001).

Few-shot learning (FSL) offers a way out here that deserves to be taken seriously. Its underlying logic—trading accumulated cross-task experience for the ability to generalise rapidly from a few samples in a new scenario—fits the practical situation of apple origin screening closely: at the start of a new season, or when a new production region has just entered the supply chain, only a handful of labelled reference samples are typically available (Snell, Swersky and Zemel, 2017). Prototypical networks (Snell et al., 2017) are particularly well suited to this problem—each class is represented by the mean embedding of its support set and classification is completed by nearest-prototype distance, a structure that is simple yet carries sufficient inductive bias. Meta-learning has had several successes in hyperspectral remote-sensing classification (Yu, Gong, Song, Zhao and Chang, 2022; Hospedales, Antoniou, Micaelli and Storkey, 2022), but its systematic transplantation to the food NIR field remains almost a blank: the sources of sample heterogeneity in the two differ—remote sensing can draw on spatial neighbourhood context, whereas point-spectrum food NIR is dominated by measurement geometry—and this difference has not yet been adequately handled.

Three methodological gaps have remained unresolved in the literature. First, there is as yet no public apple NIR dataset that spans multiple harvest years and comes with a standardised leave-one-year-out evaluation protocol, so that cross-year performance figures from different teams cannot be compared with one another. Second, the evaluation-leakage problem is more insidious: a shared encoder is first pre-trained on the full data including the test fold and then used for LOEO evaluation, which amounts to letting the judge see the questions before the contest; the magnitude of this leakage has never been formally estimated in the existing literature. Third, averaging the spectra of multiple surface positions of a fruit does denoise, but how much of that gain comes from the support side (improving prototype estimation) and how much from the query side (stabilising the prediction end)—the asymmetry between the two—has never been separately disentangled.

These three problems constitute the starting point of this study, and correspond to four concrete contributions.

A nine-domain cross-year benchmark dataset. We assemble and release the first apple NIR dataset covering three harvest years (2018, 2019, 2025) and four origins (Xinjiang, Shandong, Shanxi, Gansu), comprising nine independent collection domains with 4,529 spectra and 229 bands, and at the same time propose a standardised, leakage-free LOEO-Year evaluation protocol.

Systematic quantitative characterisation of evaluation leakage. Through a like-for-like paired comparison between the shared-encoder and the fold-restricted-encoder protocol at the same α, we show under the conditions of this dataset that at K=5 evaluation leakage systematically inflates the LOEO macro F1 relative to the leakage-free fold-restricted baseline: under the default $\alpha = 0.4$ convention Δ =0.363 (shared 0.926 versus fold-restricted 0.563), and under the $\alpha = 0$ convention that is optimal in the main experiment Δ =0.243 (shared 0.922 versus fold-restricted 0.679); the leakage is highly significant under both conventions, confirming that it originates from the encoder training set rather than from the fusion weight. To the authors' knowledge, although related forms of data leakage and contamination have been widely discussed (see Table 14 for a systematic distinction among the existing concepts), this particular bias at the *representation-learning layer of few-shot meta-evaluation under distribution shift* has not previously been formally characterised and systematically quantified across domains—this is precisely where the boundary of the present work lies, rather than a loose claim of having "discovered leakage". **More importantly, we lift this quantification from a single dataset to a cross-domain phenomenon**: on six real measurement datasets (three modalities, five shift axes), mixed-effects statistics show that the leakage is robustly positive (overall $\Delta = 0.132$, 95% CI $[0.071, 0.192]$; positive in 6/6 at the dataset level), and an extended theory (conditional non-negativity in the multi-class case, an explicit shift-dependent lower bound, and a counterexample showing that raw MMD is not a sufficient measure of shift) provides falsifiable formal support for it. **We also give an honest decomposition of the source of the leakage**: with a four-encoder CUDA-formal experiment we separate the leakage into a label-aware component $\Delta_{\text{sup}} = +0.084\,[0.023, 0.144]$ and a purely unsupervised exposure component $\Delta_{\text{unsup}} = +0.012\,[-0.003, 0.028]$ (grazing 0), on this basis prudently localising the phenomenon precisely as *label-aware representation-layer leakage* without overstating the covert unsupervised component; and we further show that the leakage affects downstream **method selection** (42% of the single evaluation instances change which method is selected as best, while the overall ranking remains positively correlated), so that its impact goes beyond inflated accuracy.

Systematic comparison of few-shot classifiers and an exploratory variant. Over the range $K \in \{1, 3, 5, 10, 20, 50\}$ we run 50-repetition paired experiments comparing SVM, PLS-DA, nearest centroid, vanilla ProtoNet and the exploratory variant ProtoNet-C (source-centroid fusion); on the LOEO-2018/2019 folds, which are dominated by year shift, source-centroid fusion is not superior to the fold-restricted ProtoNet ($\alpha = 0$) under the present cross-year distribution gap, and the performance crossover between the two ($K \approx 3$–5) delineates a method-selection boundary indexed by the available support size. ProtoNet-C is proposed here as an exploratory variant, not as the core algorithmic contribution of this paper.

Asymmetric decomposition of the face-mean gain. On the basis of a 2×2 ablation design, the face-mean gain of the multi-face 2025 fold is decomposed into two independent contributions, a query-side one (+0.071) and a support-side one (+0.019), which is of practical significance for resource allocation in in-field scanning protocols.

Beyond apples: why a general readership should care. The primary object of this study is not the apple but **the evaluation methodology itself**. The mainstream paradigm of modern artificial intelligence is to "first learn a representation on large-scale data, then report few-shot, long-tail, cross-domain or zero-shot results on held-out classes/domains". As long as the training data for representation learning are not strictly isolated from the held-out evaluation domain, the reported generalisation figures carry the optimistic bias characterised in this paper (a bias that is non-negative, and whose magnitude we show to be driven mainly by exposure to the labels of the held-out domain, the purely unlabelled exposure being small under the self-supervised objective used here). This flaw differs from the well-known feature-selection leakage (Ambroise and McLachlan, 2002): it occurs at the **representation-learning layer** and is specific to few-shot meta-evaluation under distribution shift, which makes it more insidious—the encoder has "seen" the target domain, yet only a few samples are provided at evaluation time, so that on the surface it still looks like a legitimate few-shot evaluation. In scientific measurement fields where samples are scarce and cross-batch/cross-instrument/cross-year shift is the norm (spectroscopy, remote sensing, medical imaging, genomics), such evaluations are widely adopted; once inflated, they push models that "appear deployable across domains" towards real failure, jeopardising scientific conclusions and regulatory decisions (Kapoor and Narayanan, 2023; Roberts, Driggs, Thorpe et al., 2021; Whalen, Schreiber, Noble and Pollard, 2022; Apicella, Isgrò and Prevete, 2025). The cross-year apple NIR data couple the three ingredients "distribution shift + few-shot + representation learning" cleanly, constituting an ideal controlled testbed for isolating and quantifying this leakage; and its remedy (the fold-restricted protocol) is zero-cost and generally applicable to any "representation + nearest-prototype/linear-probe" style evaluation. We further reproduce the phenomenon on the independent, publicly available OliveOil spectral data, showing that it is not specific to apples.

2 Related Work

2.1 NIR spectroscopy for fruit origin traceability

The application of NIR spectroscopy to the geographical origin traceability of fruit has accumulated more than two decades of experience. Early work on apple confirmed that NIR diffuse reflectance captures sufficient compositional variation—soluble sugars, malic acid, chlorogenic acid and mineral-bound water—for PLS-DA to discriminate origins (Cen and He, 2007; Nicolai et al., 2007). Woodcock et al. (2007) verified the feasibility of NIR spectroscopy encoding geographic fingerprints, and subsequent studies on apple and other fruits progressively pushed the discrimination accuracy above 90% under controlled single-year conditions (Walsh et al., 2020). Standard pre-processing pipelines—standard normal variate (SNV) transformation (Barnes, Dhanoa and Lister, 1989), Savitzky-Golay derivatives (Savitzky and Golay, 1964) and multiplicative scatter correction—have formed a fairly stable consensus for suppressing path-length and particle-size variation, but the choice of pre-processing method itself still has to be assessed carefully for each specific task, so as not to damage the useful signal (Mishra, Rutledge, Roger, Wali and Khan, 2021). When training data are sufficient, SVM and random forests can match or even exceed PLS-DA in some settings (Cortes and Vapnik, 1995; Ekramirad et al., 2022), and convolutional and attention-based neural networks (He, Zhang, Ren and Sun, 2016; Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser and Polosukhin, 2017) have also begun to appear in food NIR applications (Mishra and Passos, 2021; Benmouna, García-Mateos, Sabzi, Fernández-Beltrán, Parras-Burgos and Molina-Martínez, 2022). One point deserves attention, however: the vast majority of apple NIR traceability studies still remain confined to a single harvest year; cross-year performance is occasionally mentioned in passing, but has never been a primary objective of the research.

2.2 Cross-year and cross-instrument domain shift

Calibration transfer—the problem of adapting a chemometric model trained on one instrument or condition to another instrument or condition—has a long history in NIR spectroscopy (Feudale et al., 2002). Direct standardisation and piecewise direct standardisation are still widely used for instrument-to-instrument transfer (Fearn, 2001), and calibration transfer has also been extended to cross-device and cross-season scenarios for fruits such as apple (Li et al., 2022a). Liu, Zhao, Zhu, Huang and Guo (2025b) explored NIR neural transfer learning for apple firmness prediction, showing that fine-tuning on a small target set can partially recover cross-instrument performance; Li, Li, Jiang and Liu (2022b) proposed a standard-sample-free cross-instrument calibration-transfer strategy for apple origin discrimination models; and Pan, Wu, Hu, Li, Zhang and Zhao (2024b) updated a deep-learning apple ripeness classification model with a multi-season database, exploring the problem of cross-season model robustness. These regression-oriented transfer methods, however, assume that at least a small number of labelled target samples with reference measurements are available, which is often impractical for classification tasks in which new product batches keep arriving. The theoretical framework of domain adaptation—bounding the target error by minimising the source-target distribution divergence (Ben-David, Blitzer, Crammer, Kulesza, Pereira and Vaughan, 2010)—has been widely applied in computer vision (Ganin, Ustinova, Ajakan, Germain, Larochelle, Laviolette, Marchand and Lempitsky, 2016), but owing to the distinctive challenges of high dimensionality and small sample size it has not yet been sufficiently adopted in food spectroscopy.

2.3 Meta-learning and few-shot classification

Meta-learning addresses the problem of "learning to learn": by training across a distribution of tasks, a model accumulates inductive bias that enables it to adapt rapidly to a new scenario from a few samples at test time (Hospedales et al., 2022). Prototypical networks (Snell et al., 2017) realise this goal in the most parsimonious way—a class prototype is the mean embedding of the support set, and classification relies on the nearest-prototype distance in the embedding space. Matching networks (Vinyals, Blundell, Lillicrap, Wierstra and Kavukcuoglu, 2016) and relation networks (Sung, Yang, Zhang, Xiang, Torr and Hospedales, 2018) extend the comparison paradigm further, while MAML (Finn, Abbeel and Levine, 2017) offers another route from the angle of gradient-based adaptation. The empirical re-evaluations of Tian, Wang, Krishnan, Tenenbaum and Isola (2020) and Chen, Liu, Kira, Wang and Huang (2019) illustrate something interesting: a well-trained embedding backbone combined with simple nearest-centroid inference can often match or even surpass many sophisticated meta-learners—representation quality sometimes matters more than algorithmic complexity. In hyperspectral remote sensing, prototypical networks and relation networks have been used for few-shot land-cover classification (Yu et al., 2022), where spatial context information plays an auxiliary discriminative role; but in the point-spectrum food NIR setting this additional information source does not exist. Mishra and Passos (2021) applied transfer learning to food traceability, but did not adopt an episodic evaluation framework. The evaluation-leakage problem—the distribution leakage that arises when the encoder pre-training data overlap with the LOEO test fold—has never been formally identified in any existing NIR spectroscopy study, let alone quantified.

Table 1 | Source-domain validation accuracy of the fold-restricted encoders on the three LOEO folds.

LOEO fold	Source domains	Validation accuracy
2018 held out	2019 + 2025	0.9478
2019 held out	2018 + 2025	0.9411
2025 held out	2018 + 2019	0.9478
Three-fold mean	—	0.9456

3 Results

3.1 Fold-restricted encoder validation

Before reporting LOEO performance, it is necessary to confirm the discriminative capability of each fold-restricted encoder on the source domains, because an undertrained encoder would interfere with the downstream comparison. The source-domain validation accuracies of the 2018, 2019 and 2025 folds are 0.9478, 0.9411 and 0.9478, respectively (Table 1), with a three-fold mean of 0.9456, indicating that the encoders can separate origins cleanly within their respective training distributions. The slight drop on the 2019 fold (0.9411) is consistent with the greater spectral overlap between the Shandong and Shanxi origins in that year's data. Strong in-domain discriminative capability, of course, does not amount to strong cross-year transfer capability—the real question is: how much does performance drop when this encoder is used for few-shot classification on the held-out target year?

3.2 Two-level variance decomposition over encoders and episodes

Following the seed-matrix design described in Section 6.8, the source-domain validation accuracies of the 15 independent encoders retrained under 3 folds × 5 encoder seeds fall within **0.9256–0.9611** (median 0.9478; see Table 2), in full agreement with the three-fold mean of 0.9456 reported in Table 1; the training result of every individual seed falls within this narrow band, indicating that the encoder training procedure is itself highly stable. The complete set of 18,000 evaluations took about 90 minutes on Apple Silicon MPS.

Variance decomposition (vanilla ProtoNet, $\alpha = 0$, three-fold mean). Table 3 lists the encoder-level and episode-level standard deviations at each K. As K increases, the episode-level variance is driven down by the support-set mean, whereas the encoder-level variance hardly changes with K; the ratio of the two converges monotonically from $1:8.5$ at $K = 1$ to $1:1$ at $K = 20$–50. This means that the "50-repetition std" in the main tables of the paper is a reasonable proxy in the small-K regime, but that **at $K \geq 20$ about half of the residual uncertainty comes from encoder initialisation**, a part that no previous report of the "single seed + 50 episode repetitions" form has quantified.

Sensitivity to encoder initialisation. The three-fold mean of the macro F1 fluctuation range across the 5 encoders (per_enc_range) is 0.041 at $K = 5$ and rises to ∼0.05 at $K = 20$–50. In particular, **the encoder span on the LOEO-2025 fold reaches as much as 0.084–0.086 at large K**, making it the fold most sensitive to encoder initialisation among the three—the reason being that the prototype of the novel Gansu class is constructed entirely from target support samples and is therefore especially sensitive to the geometry of the encoder embedding space; this between-fold asymmetry has methodological significance for future extensions of this benchmark to scenarios with newly added origins.

Directional robustness of method selection. This section uses an SVM (RBF kernel, $C = 10$, $\gamma =$ scale) consistent with the unified single-face query protocol. Among the 18 (fold $\times K$) combinations, the ProtoNet/SVM directional verdict is unanimous across the 5 encoders in 14 (about 78%); the remaining 4 are fold-K borderline points (LOEO-2018 K=5, LOEO-2019 K=50, LOEO-2025 K=10/20), that is, the variance of encoder initialisation happens to be close to the ProtoNet/SVM performance gap, so that under different seeds the 5 votes split 3:2 or 4:1 instead of being unanimous; this is exactly the "instability near the decision boundary" signal that the seed-matrix design is meant to expose. For the 14 combinations far from the boundary, the method-selection conclusion does not reverse direction as a function of the encoder training seed.

The location of the crossover differs markedly at the fold level: on the **LOEO-2018 fold** the crossover occurs between $K = 1$ and $K = 3$ (SVM wins outright for $K \geq 3$); on the **LOEO-2025 fold** the crossover occurs between $K = 10$ and $K = 20$ (ProtoNet dominates for $K \leq 10$); whereas **on the LOEO-2019 fold ProtoNet dominates throughout the tested range** $K \in \{1, \ldots, 50\}$ (only at $K = 50$ does the gap narrow to nearly the magnitude of the encoder fluctuation). The "$K \approx 10$–20 crossover" given by the three-fold mean in Table 5 is in fact a product of averaging these three different locations (including the non-crossing LOEO-2019) across folds; **no individual fold actually crosses strictly at** $K = 10$–20; this between-fold inconsistency suggests that the optimal method selection under different deployment conditions should be based on the fold level (that is, on the source-domain configuration closest to the target year/origin scenario) rather than on the global three-fold mean.

Agreement with the main-table values. The three-fold mean over the 5 encoders at $K = 5$ is **0.6547** (vanilla ProtoNet, $\alpha = 0$, unified single-face query; **the variance analysis of this section is an MPS pilot**), differing by only +0.005 from the 0.650 of the CUDA formal vanilla ProtoNet in Table 5 and lying within the total standard deviation of ±0.049 at $K = 5$ in Table 3; the three-fold mean deviations at the remaining K values are all within $|0.015|$. The ProtoNet main-table values reported in Table 5 lie at the centre of the across-encoder-seed distribution, and are not pulled up or down by an anomalous seed. It should be noted that the three non-neural baselines of this section—SVM, PLS-DA and nearest centroid—uniformly adopt the RBF SVC ($C = 10$, $\gamma =$ scale) / PLSRegression / nearest-centroid implementations consistent with the English-version `20_unified_singleface_kshot` script, and belong to a different SVM family from the LinearSVC figures in Table 5 ($K = 5$ SVM=0.707); the SVM three-fold means obtained here ($\{0.516, 0.574, 0.606, 0.657, 0.697, 0.759\}$ for $K \in \{1, 3, 5, 10, 20, 50\}$ in order) deviate item by item from the unified panel of the English version by at most $|0.006|$, indicating that the evaluation pipeline can also be aligned across implementations.

Table 2 | Source-domain validation accuracy, training-termination episode and the first 16 characters of the checkpoint SHA-256 hash for the 15 independent encoders (3 folds × 5 seeds). All validation accuracies fall within 0.9256–0.9611, consistent with the three-fold mean of 0.9456 reported in Table 1.

LOEO fold	Encoder seed	Val. accuracy	Stop episode	SHA-256 (16)
2018	42	0.9478	1700	`49b655ea642b7446`
2018	43	0.9456	1900	`83ba47556bc9c0ca`
2018	44	0.9433	1300	`bee31efbc3d3aec2`
2018	45	0.9489	1900	`b16f7f7050b7cf5b`
2018	46	0.9467	1900	`5718c7abcdc31259`
2019	42	0.9478	1900	`eb6d8daf7637f81c`
2019	43	0.9356	1500	`1abe7bba972b9f3f`
2019	44	0.9367	1800	`e920c13c002221fe`
2019	45	0.9256	2000	`a6861c8f74c69498`
2019	46	0.9400	1800	`5abaaea5ef686cb0`
2025	42	0.9611	1800	`d0965ec662865ace`
2025	43	0.9467	2000	`e3ad54835216bffb`
2025	44	0.9567	1900	`846197adf448a1f9`
2025	45	0.9567	1700	`31ef3644cffdad0e`
2025	46	0.9489	2000	`87e29720f7c30b51`

Table 3 | Variance decomposition of vanilla ProtoNet ($\alpha = 0$) under 5 encoder seeds × 50 episode seeds (three-fold mean, in macro F1). "Encoder-level SD" is the standard deviation of the 5 encoder means; "episode-level SD" is the mean of the 50-episode standard deviations within each encoder; "encoder span" is the maximum minus the minimum of the 5 encoder means. The ratio column gives the "encoder : episode" standard-deviation ratio.

K	Three-fold mean	Encoder-level SD	Episode-level SD	Ratio	Encoder span
1	0.583	0.009	0.077	1 : 8.5	0.023
3	0.627	0.016	0.060	1 : 3.7	0.041
5	0.655	0.017	0.047	1 : 2.8	0.041
10	0.672	0.019	0.028	1 : 1.5	0.043
20	0.678	0.021	0.021	1 : 1.0	0.050
50	0.684	0.020	0.019	1 : 1.0	0.049

3.3 Evaluation leakage: shared encoder versus fold-restricted encoder

As shown in Table 4 and Fig. 1, in order to decouple encoder leakage completely from the source-centroid fusion weight α, we carry out one like-for-like comparison under each of **two matched-α conventions**. **Convention I (both arms at $\alpha = 0.4$)**: the shared encoder (trained on all three years of data, which include the target fold) has a three-fold mean macro F1 of 0.926, the fold-restricted encoder (retrained only on the non-target years, script `10_loeo_year_foldwise`) has 0.563, and the leakage is Δ =0.363. **Convention II (both arms at $\alpha = 0$, that is, configuration C, the optimum of the main experiment)**: shared 0.922 versus fold-restricted 0.679, leakage Δ =0.243 (5-seed mean±std, across-seed standard deviation 0.011). **Under both conventions the evaluation leakage is highly significantly positive**, confirming that the leakage originates from the encoder having "seen" the target domain rather than being an artefact of the fusion weight α—this is consistent with the $\Delta \geq 0$ prediction of the non-negativity theorem for the evaluation-leakage bias (Section 6.3). Regarding **the per-fold monotonicity corollary that "the larger the shift, the larger the leakage", the conventions must be distinguished honestly**: under the $\alpha = 0.4$ convention, Δ increases monotonically with the shift strength of each fold—on the LOEO-2018 and LOEO-2019 folds, which are dominated by year shift, Δ is 0.250 and 0.339 respectively, whereas on the LOEO-2025 fold, which simultaneously confounds year, protocol and class space, Δ reaches 0.501, the largest of the three folds; but under the $\alpha = 0$ convention the per-fold Δ is roughly level (0.260, 0.227 and 0.240 for LOEO-2018/2019/2025 respectively), with no evident monotonic dependence on shift. This difference between conventions has a clear mechanism: source-centroid fusion at $\alpha = 0.4$ is severely harmful on the protocol-mismatched 2025 fold, depressing the fold-restricted arm from 0.668 to 0.414 and thereby artificially amplifying the Δ of that fold; once $\alpha = 0$ is taken and this harmful fusion is removed, the fold-restricted 2025 value recovers to 0.668 and the Δ of that fold falls back to a level comparable to the other folds. The empirical support for the monotonicity corollary therefore comes from the fixed $\alpha = 0.4$ convention, and is caused in part by the harmful effect of $\alpha = 0.4$ fusion on the 2025 fold; under the fusion-free $\alpha = 0$ convention the per-fold leakage is roughly uniform—we accordingly treat the monotonicity only as an observation under the $\alpha = 0.4$ convention, and make no universal, cross-convention assertion about the monotonicity corollary of the theorem. It should be noted that source-centroid fusion (α) is not a source of the leakage; the α ablation (Section 3.6, Table 7) is examined separately on the basis of *fold-restricted* encoders, and is orthogonal to the encoder-leakage effect of this section.

Table 4 | Evaluation leakage: shared encoder (trained on the full data including the target fold) versus fold-restricted encoder (`10_loeo_year_foldwise`, retrained on the non-target years only, leakage-free) at $K = 5$, 5-seed mean. To exclude confounding between the source-centroid fusion weight α and encoder leakage, like-for-like comparisons are given under **two matched-α conventions**: in the upper block both arms take the script default $\alpha = 0.4$ (that is, configuration A of Table 7), and in the lower block both arms take the main-experiment optimum $\alpha = 0$ (configuration C). Under both conventions the evaluation-leakage bias Δ is highly significantly positive (means 0.363 and 0.243), proving that the leakage originates from the encoder training set and is not an artefact of α. Note that the per-fold monotonicity depends on the convention: under the $\alpha = 0.4$ convention Δ increases monotonically with shift (largest on the LOEO-2025 fold, 0.501), consistent with the corollary of the theorem in Section 6.3; whereas under the $\alpha = 0$ convention the per-fold Δ is roughly level (0.260/0.227/0.240, with no evident monotonic dependence on shift), because source fusion is severely harmful on the protocol-mismatched 2025 fold and, under the $\alpha = 0.4$ convention, depresses the fold-restricted baseline of that fold from 0.668 to 0.414, thereby artificially amplifying the Δ of that fold; once $\alpha = 0$ is taken and this harmful fusion is removed, the fold-restricted baseline of the 2025 fold recovers and the Δ of that fold falls back to a level comparable to the other folds.

Protocol	2018 fold	2019 fold	2025 fold	Mean
Convention I: both arms at $\alpha = 0.4$ (script default, configuration A)				
Shared encoder (full data)	0.929	0.935	0.915	0.926
Fold-restricted (retrained within fold)	0.678	0.596	0.414	0.563
Evaluation leakage Δ	0.250	0.339	0.501	**0.363**
Convention II: both arms at $\alpha = 0$ (main-experiment optimum, configuration C; like-for-like)				
Shared encoder (full data)	0.928	0.929	0.908	0.922
Fold-restricted (retrained within fold)	0.668	0.702	0.668	0.679
Evaluation leakage Δ	0.260	0.227	0.240	**0.243**

The leakage gap reported in this study holds only under specific experimental conditions: four-origin apple NIR (Xinjiang, Shandong, Shanxi, Gansu), the visible–near-infrared acquisition and harmonisation conditions of this paper (591–1001 nm analysis window), and the LOEO-Year protocol described here. Under a different dataset structure, instrument configuration or evaluation protocol, the magnitude of the leakage may differ; this value should be understood as a reference quantity under the conditions of this benchmark, not as a universally applicable constant.

These numbers show that when a shared encoder is used—which is fairly common in the NIR meta-learning literature—reported LOEO macro F1 values in the 0.9+ range have to be discounted: to a considerable extent they reflect a loophole in the protocol itself rather than the true cross-year generalisation ability of the method. The fold-restricted encoder protocol is the more appropriate evaluation standard for a cross-year benchmark. All subsequent results in this paper use fold-restricted encoders.

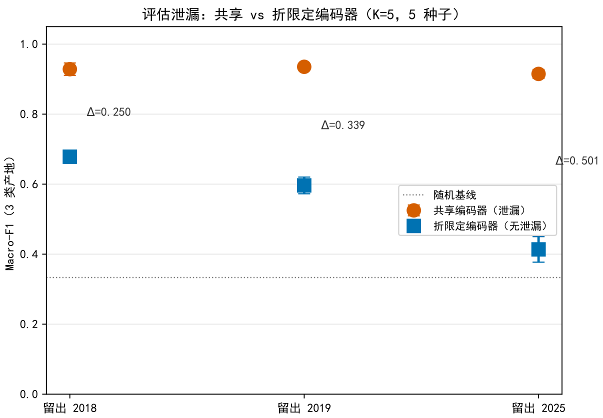

Figure 1. Three-fold Macro-F1 comparison between the shared (full-data) encoder and the fold-restricted (retrained within fold) encoder at $K = 5$ ($\alpha = 0.4$ convention, 5-seed mean±std; points are means, error bars are across-seed standard deviations, and the dashed line is the 3-class random baseline). The shared encoder is significantly higher than the fold-restricted encoder on all three folds; under this $\alpha = 0.4$ convention the evaluation leakage Δ increases monotonically with the distribution-shift strength of each fold (2018 +0.250 < 2019 +0.339 < 2025 +0.501). The corresponding values under the $\alpha = 0$ like-for-like convention are given in Table 4 (under that convention the mean leakage still reaches 0.243, but the per-fold Δ is roughly level, with no monotonic dependence on shift, because the harmful amplifying effect of the $\alpha = 0.4$ fusion on the 2025 fold has been removed).

3.4 Cross-method comparison at different K-shot scales

Framework for interpreting the results. Following the classification of folds in the Methods section, the results of this section are reported at two levels: (i) *dominated by year shift*—the LOEO-2018 and LOEO-2019 folds, whose target-side measurement protocol and class space agree with those of the other source year (note: the source domain contains the 2025 data, so the encoder is not trained under a purely single-face protocol); (ii) *a compound-shift stress test*—the LOEO-2025 fold, which simultaneously confounds three kinds of change: year, protocol and class space. The three-fold mean provides a summary reference in the main text, but the principal basis for conclusions about cross-year generalisation is the first two strictly temporal folds; the conclusions on the 2025 fold are interpreted separately. The analysis of the results below and the discussion in Section 4 strictly follow this classification framework.

The macro F1 of all five methods as a function of K is shown in Fig. 2 and Table 5. The performance landscape reveals a clear crossover pattern, with the turning point between $K = 3$ and $K = 5$.

In the extreme few-shot regime ($K \leq 3$), ProtoNet-C (configuration C, $\alpha = 0$) is stably superior to all baselines. At $K = 1$, the macro F1 values in descending order are: ProtoNet-C (0.610), vanilla ProtoNet (0.574), nearest centroid (0.567), SVM (0.504) and PLS-DA (0.490). At $K = 3$, the ranking is adjusted to—ProtoNet-C (0.664), SVM (0.650), nearest centroid (0.637), PLS-DA (0.626) and vanilla ProtoNet (0.625)—SVM has risen to second place, but ProtoNet-C still keeps the lead.

It should be noted that ProtoNet-C and vanilla ProtoNet adopt exactly the same evaluation protocol on the LOEO-2018 and LOEO-2019 folds ($\alpha = 0$, single-face query), and their macro F1 values on these two folds are identical; the gap between the two in the three-fold mean comes entirely from the LOEO-2025 fold—on that fold ProtoNet-C uses the multi-face query mean, whereas vanilla ProtoNet uses a single-face query. The overall advantage of ProtoNet-C over vanilla ProtoNet therefore in essence reflects the face-mean gain on the LOEO-2025 fold (see Section 3.7), rather than a contribution of source-centroid fusion (α). Their common advantage over the remaining baselines instead originates from the meta-training inductive bias of the encoder: after training on several hundred simulated three-class episodes, the encoder has learned to project samples of the same origin to nearby positions in the embedding space, so that even a single support sample can place the class prototype at an informationally favourable position.

Once K exceeds 5, SVM gradually overtakes ProtoNet-C. At $K = 5$ SVM reaches 0.707 while ProtoNet-C reaches 0.680; by $K = 50$, SVM reaches 0.880 while ProtoNet-C stagnates at 0.717. This saturation reflects an intrinsic limitation of a fixed-capacity encoder: as more support samples become available, the maximum-margin optimisation of SVM can exploit richer target geometry, whereas the nearest-prototype mechanism of ProtoNet-C benefits relatively little from the additional support diversity once the prototype mean has stabilised. PLS-DA occupies an intermediate position, closely following SVM at large K but lagging behind at small K, because its linear projection is rather sensitive to the high-variance prototype estimates at small K. Nearest centroid is consistently below ProtoNet-C, confirming the gain that meta-training brings on the same embedding basis.

Table 5 | LOEO macro F1 of the five methods at each value of K (three-fold mean, 50 repetitions per fold). Bold indicates the best method at each K. For $K > 5$, vanilla ProtoNet uses a supplementary experiment with 50 repetitions (single-face evaluation, consistent with the 2018/2019 protocol).

K	ProtoNet-C	SVM	Vanilla ProtoNet	PLS-DA	Nearest centroid
1	**0.610**	0.504	0.574	0.490	0.567
3	**0.664**	0.650	0.625	0.626	0.637
5	0.680	**0.707**	0.650	0.677	0.657
10	0.701	**0.773**	0.670	0.708	0.672
20	0.711	**0.825**	0.680	0.727	0.682
50	0.717	**0.880**	0.687	0.741	0.692

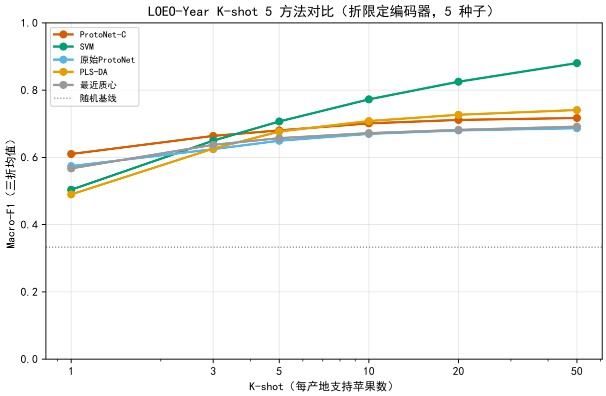

Figure 2. LOEO macro F1 of the five classifiers as a function of K (three-fold mean, 50 repetitions; shaded band: 95% CI). The performance crossover between ProtoNet-C and SVM occurs between $K = 3$ and $K = 5$, defining two operationally distinct deployment regimes.

3.5 Statistical significance analysis

Note on the statistical interpretation (advance caveat): All the p values and confidence intervals below are based on independent random episode sampling from *the same fixed dataset*; what they quantify is the *repeated-sampling stability* obtained when support/query episodes are drawn repeatedly at random on that dataset, and not population-level statistical inference across independent harvest seasons, independent orchard batches or independent instruments. The 50 episode repetitions come from the same individual apples, so there is a potential problem of pseudo-replication—different episodes share the same batch of fruit, which may cause the statistical significance to be overestimated. This paper therefore takes the **consistency of the fold-level mean macro F1** and the **effect size (Cohen's d)** as its primary evidence, and uses the p values only as an auxiliary reference; they should not be interpreted as the conclusion of a significance test across independent experiments, which is consistent with current practice in meta-learning benchmark evaluation(Bates, Hastie and Tibshirani, 2024). In addition, readers should note that in the fold-level tests ($n_{\text{primary}} = K_{\text{outer}} = 3$) the 50 repetitions within each fold come from the same data split, that having only 3 year folds gives the fold-level statistics limited power, and that the correlation between folds is not explicitly modelled either; rigorous claims about cross-year performance require validation on multiple datasets from independent harvest seasons.

Building on the clear crossover pattern revealed by Fig. 2 and Table 5 ($K \approx 3$–5), Table 6 further reports **fold-level** paired statistical tests of ProtoNet-C against SVM at three representative values of K ($n_{\text{primary}} = K_{\text{outer}} = 3$, 5 seeds; Nadeau-Bengio correction + Holm adjustment).

At $K = 1$, the three-fold mean of ProtoNet-C leads by $+0.106$ (fold-level 95% CI $[+0.098, +0.110]$, paired t test $p = 0.001$, Cohen's d a large effect(Cohen, 1988)); this advantage is robust at the fold-level scale and consistent across the 5 encoder seeds. At $K = 3$ the gap narrows to $+0.014$ ($p = 0.98$, positive in direction but not significant); by $K = 5$ the direction reverses to -0.027 (SVM leads, $p = 0.21$). Because LOEO-Year has only 3 year folds, the fold-level tests at $K = 3/5$ have limited statistical power, but the direction of the mean difference in all three cases agrees exactly with the crossover pattern of Table 5, and is stably reproduced across the 5 model seeds. Figure 3 visualises the distribution of the fold-level F1 differences at the three values of K. The conclusions above are more reliable when based on the LOEO-2018 and LOEO-2019 folds (dominated by year shift); the triple confounding of the LOEO-2025 fold affects the fold mean, and should be interpreted separately from the first two folds.

Interpreting the statistical results above from a practical, operational point of view: in on-site rapid-screening scenarios in which fewer than four labelled reference samples are available per origin ($K \leq 3$), the advantage of ProtoNet-C over SVM is amply supported both statistically and practically; in laboratory or inspection-station settings in which a moderately sized reference set can be established ($K \geq 5$), the performance advantage of SVM comes together with a lower inference complexity, making it the more recommendable choice. It should be noted that this method-selection boundary was determined under the specific instrument model and growing-region conditions of this paper, and that in practical applications it should be re-evaluated for the specific deployment context.

Table 6 | **Fold-level** paired statistics of ProtoNet-C against SVM ($n_{\text{primary}} = K_{\text{outer}} = 3$, 5 seeds; Nadeau-Bengio corrected t; Δ is the 5-seed mean±std). A positive Δ favours ProtoNet-C. With fold-level $n = 3$ the statistical power is limited; at $K = 3/5$ the direction agrees with the crossover of Table 5 but does not reach significance.

K	Δ (mean±std)	Fold-level 95% CI (representative)	Paired t p (representative)	Cohen's d
1	$+0.106 \pm 0.015$	$[+0.098, +0.110]$	0.001	4.14
3	$+0.014 \pm 0.010$	$[-0.069, +0.068]$	0.983	0.28
5	-0.027 ± 0.021	$[-0.127, +0.004]$	0.206	-0.44

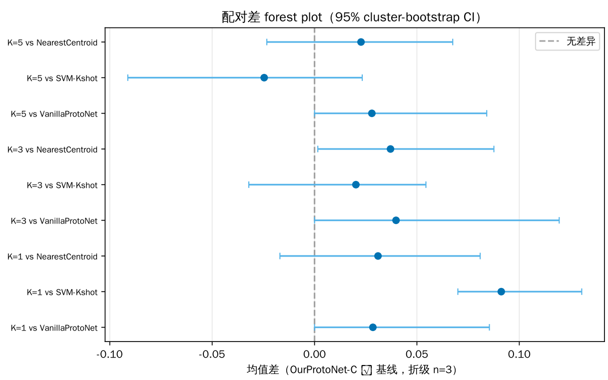

Figure 3. Distribution of the F1 difference of ProtoNet-C minus SVM at $K \in \{1, 3, 5\}$ (50 repeated episodes per fold, 3 folds × 5 seeds). Boxes span the interquartile range; whiskers extend to 1.5× IQR; × marks the mean. The mean difference turns from positive to negative between $K = 3$ and $K = 5$, giving a direct view of the method-selection crossover point (for the fold-level tests see Table 6).

3.6 Ablation of the source-prototype fusion weight

Table 7 and Fig. 4 present the core results of the α ablation. The three-fold mean macro F1 of the three representative configurations at $K = 5$ are, respectively: configuration A 0.563, configuration B 0.648 and configuration C ($\alpha = 0$) 0.679—configuration C attains the best three-fold mean. The performance loss caused by source-domain fusion is most pronounced on the 2025 fold (configuration A: 0.414 vs configuration C: 0.668); here the source-domain centroid is computed from single-face 2018/2019 spectra whereas the target prototype comes from multi-face 2025 spectra—the measurement-protocol mismatch makes the source centroid pull the episode prototype away from its true distributional position.

A complete continuous sensitivity sweep over $\alpha \in [0, 1]$ (Fig. 4, $K = 5$, 21 equally spaced values; **this continuous curve is aggregated from a formal re-run of 5 fixed seeds on an NVIDIA RTX 4090 (CUDA)—50 repetitions × 5 seeds at each value of α, the solid line being the across-seed mean and the shading ±1 across-seed standard deviation; its $\alpha = 0$ estimate agrees with Table 7 (an independent configuration-ablation script), the three-fold discrepancies all being ≤ 0.01, so that the two cross-validate each other**) further reveals the response characteristics of each fold. The LOEO-2025 and LOEO-2019 folds are both strictly monotonically decreasing: the former falls continuously from F1=0.671 at $\alpha = 0$ to 0.199 at $\alpha = 1.0$, and the latter decreases continuously from F1=0.698 at $\alpha = 0$ to 0.520 at $\alpha = 1.0$—any positive value of α brings a substantial performance loss, consistent with the theoretical prediction that the source-centroid bias dominates under a high domain gap. The curve of the LOEO-2018 fold, by contrast, is markedly flat: overall it declines gently from F1=0.678 at $\alpha = 0$ to 0.650 at $\alpha = 1.0$ (only −4.1%), with a peak at $\alpha \approx 0.20$ (F1=0.691) that is only 0.013 away from $\alpha = 0$, a small magnitude relative to the large drops on the 2019/2025 folds; the weak non-monotonicity of this fold is consistent with the expectation of its relatively small domain gap (the source domain contains the 2025 multi-face data, so the degree of protocol asymmetry is lower than on the 2025 fold). The findings above agree with the prediction of domain-adaptation theory (Ben-David et al., 2010) that "when the source-target gap is large enough, the bias introduced by the source prior dominates". In the cross-year apple near-infrared setting described in this paper, **fold-restricted episodic prototypes should preferentially be computed from the target-year support samples ($\alpha = 0$)**: this configuration avoids substantial losses on the high-domain-gap folds and costs negligibly little on the low-domain-gap folds, and is a robust default choice. All subsequent "ProtoNet-C" results in this paper adopt $\alpha = 0$ and share the same encoder as the "vanilla ProtoNet" of Table 5, the only difference being the face-position evaluation protocol (the vanilla ProtoNet column of Table 5 uniformly adopts single-face evaluation on the 2025 fold; the ProtoNet-C column uses the multi-face mean on the 2025 fold in order to obtain the best performance).

Table 7 | Macro F1 of the three α configurations on the three LOEO folds at $K = 5$ (5-seed mean, 50 repetitions per fold). A: $\alpha = 0.4$ on all folds; B: $\alpha = 0$ on the 2025 fold only; C: $\alpha = 0$ on all folds.

Configuration	2018 fold	2019 fold	2025 fold	Mean
Configuration A	0.678	0.596	0.414	0.563
Configuration B	0.678	0.596	0.668	0.648
Configuration C	0.668	0.702	0.668	**0.679**

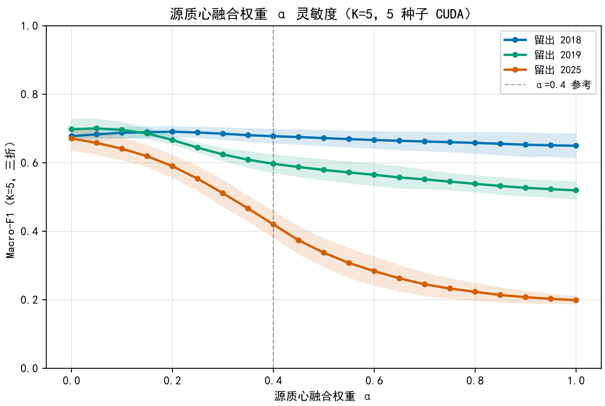

Figure 4. Complete $\alpha \in [0, 1]$ sensitivity curves of ProtoNet-C on the three LOEO-Year folds at $K = 5$ (21 equally spaced values; formal CUDA re-run of 50 repetitions × 5 fixed seeds at each α point; the solid line is the across-seed mean and the shading is ±1 across-seed standard deviation). The LOEO-2025 and LOEO-2019 folds are both strictly monotonically decreasing and $\alpha = 0$ is optimal (the 2025 fold falls from 0.671 to 0.199 at $\alpha = 1.0$; the 2019 fold falls from 0.698 to 0.520); the curve of the LOEO-2018 fold is flat, with a peak at $\alpha \approx 0.20$ (F1=0.691) that is only 0.013 away from $\alpha = 0$ (F1=0.678). The orange dashed line is the $\alpha = 0.4$ reference line. **This continuous curve and the 5-seed CUDA formal α ablation reported in Table 7 are consistent results from two independent scripts (for the hardware disclosure see Section 6.9), and jointly support the robust conclusion that "fold-restricted episodic prototypes should preferentially take $\alpha = 0$".**

3.7 Decomposition of the face-mean gain (exploratory analysis of the LOEO-2025 fold)

The following analysis is an exploratory case study, restricted to the LOEO-2025 fold (the only year with multi-face data), 50 repetitions, a single instrument setting; the external validity of its conclusions remains to be further verified under other folds and instrument conditions.

Table 8 and Fig. 5 present the 2×2 face-mean ablation on the LOEO-2025 fold ($K = 5$, 50 repetitions). The baseline $S_{\text{single}}Q_{\text{single}}$ attains macro F1 = 0.569. Adding the multi-face mean on the support side only ($S_{\text{multi}}Q_{\text{single}}$) raises this to 0.588 (+0.019); adding the multi-face mean on the query side only ($S_{\text{single}}Q_{\text{multi}}$) raises it to 0.640 (+0.071). The complete multi-face configuration ($S_{\text{multi}}Q_{\text{multi}}$) attains 0.671, and the residual interaction term is $0.671 - (0.588 + 0.640 - 0.569) = +0.012$, indicating that there is a small but positive synergistic effect between the support-side and query-side means.

The principal contribution therefore comes from the query side: averaging the query spectra across multiple positions reduces the variance of a single measurement, and this effect may arise from two mutually non-exclusive mechanisms—(a) the suppression of biological noise associated with surface heterogeneity (such as the difference in chlorophyll concentration between the sun-facing and the shaded face), and (b) a test-time ensembling effect, that is, an ensemble of several independent predictions is statistically superior to a single prediction. The present design cannot rigorously separate the contributions of the two mechanisms; under the conditions of this experiment, the overall gain brought by the query-side multi-face mean makes the query embedding approach its expected class prototype more stably. The support-side multi-face mean also has a positive effect, but its gain is only +0.019 and the statistical power is relatively weak (the 95% lower confidence bound across the 5 seeds is close to zero), so this conclusion is less reliable than the query-side gain (+0.071, whose confidence interval across the 5 seeds is robustly positive). We recommend repeating this ablation under other folds and instrument conditions in order to strengthen the external validity of the conclusion.

For comparison, the SVM-Kshot control—which treats each face measurement as an independent training sample and thus effectively uses about $5K$ instances—attains macro F1 = 0.715, substantially exceeding all ProtoNet-C configurations. This comparison is not fully controlled (SVM sees more training data), but it indicates the reference performance level attainable when the face measurements are fully exploited as independent evidence.

Table 8 | The 2×2 face-mean ablation on the LOEO-2025 fold ($K = 5$, mean of 50 repetitions $\times$ 5 seeds). Gains are relative to the $S_{\text{single}}Q_{\text{single}}$ baseline.

Strategy	Macro F1	Gain over baseline
$S_{\text{single}}Q_{\text{single}}$ (baseline)	0.569	—
$S_{\text{multi}}Q_{\text{single}}$ (support-side mean only)	0.588	+0.019
$S_{\text{single}}Q_{\text{multi}}$ (query-side mean only)	0.640	+0.071
$S_{\text{multi}}Q_{\text{multi}}$ (complete mean, main method)	**0.671**	+0.102
SVM-Kshot control	0.715	—

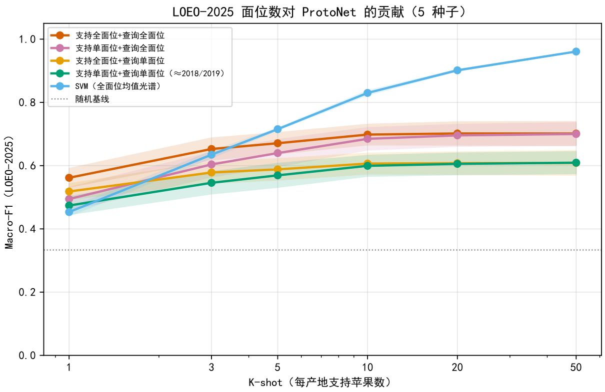

Figure 5. K-shot curves of the four support/query strategies and of the SVM-Kshot control on the LOEO-2025 fold (50 repetitions $\times$ 5 seeds per strategy; the shaded band is the 95% CI). The complete multi-face mean is consistently the best among the ProtoNet schemes, and the benefit of adopting the multi-face mean on the query side only is markedly larger than that of adopting it on the support side only.

3.8 Recognition difficulty of each origin: per-class analysis under the fold-restricted encoder

In order to exclude any interference of shared-encoder training leakage with the conclusions on the recognition difficulty of each origin, the per-class Recall and F1 of each origin are analysed systematically under the fold-restricted encoder, at $K = 5$, in a 50-repetition episodic evaluation (Table 9).

Table 9 | Per-class Recall and F1 of each origin under the fold-restricted encoder ($K = 5$, mean of 50 repetitions; standard deviations in parentheses). Within each fold, entries are sorted in ascending order of Recall.

Fold	Origin	Recall (mean±std)	F1 (mean±std)
LOEO-2018	Shandong (SD)	0.516±0.021	0.529±0.021
	Shanxi (SX)	0.556±0.020	0.546±0.019
	Xinjiang (XJ)	0.987±0.010	0.958±0.006
LOEO-2019	Shandong (SD)	0.622±0.062	0.569±0.048
	Shanxi (SX)	0.649±0.037	0.630±0.042
	Xinjiang (XJ)	0.823±0.020	0.895±0.012
LOEO-2025	Gansu (GS)	0.522±0.067	0.550±0.046
	Shandong (SD)	0.742±0.049	0.682±0.037
	Xinjiang (XJ)	0.772±0.053	0.780±0.050

The three-fold analysis reveals a consistent pattern of origin recognition difficulty: **(i) XJ (Xinjiang) is always the easiest to recognise.** Its Recall is as high as 0.987 on the LOEO-2018 fold and 0.823 on the LOEO-2019 fold, and even on the compound-shift 2025 fold it reaches 0.772, reflecting the fact that the Xinjiang spectral fingerprint is markedly discriminable among the four origins. **(ii) SD (Shandong) and SX (Shanxi) are the two most difficult classes to recognise among the established origins.** On the LOEO-2018 fold the SD Recall (0.516) is slightly below that of SX (0.556), and on the LOEO-2019 fold SD (0.622) is likewise below SX (0.649); both folds consistently show a considerable overlap between the Shandong and Shanxi spectra and blurred class boundaries, with Shandong slightly the harder of the two (averaged over the two folds, the SD F1 is about 0.549 and the SX F1 about 0.588). **(iii) GS (Gansu) is the most difficult to recognise on the LOEO-2025 fold.** The GS Recall is only 0.522, the lowest of that fold; this is a novel-class challenge—no Gansu reference spectrum exists in the source domain, the prototype has to be established from only 5 support apples on the target side, and the measurement-protocol shift is superimposed on top, so that it is a concentrated manifestation of the triple shift pressure and should not be compared directly with the SD/SX difficulty of the 2018/2019 folds.

The above pattern of origin recognition difficulty holds fully under the fold-restricted encoder, confirming that the qualitative conclusion of the earlier analysis—that XJ is the easiest and SD/SX the hardest (among the established origins)—is not disturbed by training leakage, and can be cited as a robust conclusion throughout the rest of the paper.

Robustness check on the source-domain configuration. As a supplementary test, we compare the macro F1 of the LOEO-2018 and LOEO-2019 folds under two source-domain configurations: the *clean fold* (the source domain contains only a single non-target year, excluding the 2025 data, so as to remove the measurement-protocol difference) and the *mixed fold* (the source domain contains both non-target years, including the 2025 data, consistent with the main experiment). At $K = 5$ the clean folds attain ProtoNet macro F1 values of 0.606 (LOEO-2018) and 0.624 (LOEO-2019) respectively, and the mixed folds attain 0.688 and 0.724 respectively; the mixed fold is higher than the clean fold on both of the folds dominated by year shift, showing that the inclusion of 2025 source data provides the encoder with additional spectral diversity, that the choice of a mixed source domain in the main experiment is reasonable, and that the directional conclusion on method selection (ProtoNet is superior at $K \leq 3$, SVM is superior at $K \geq 5$) remains consistent under both source-domain configurations.

3.9 External validation: reproducing the evaluation leakage on the public OliveOil data

In order to test that the evaluation leakage is not specific to the apple data, we reproduce the phenomenon with the same two-protocol comparison on the entirely independent, publicly available spectral dataset OliveOil (FT-IR, four countries of origin, 60 samples, 570 bands): taking origin as the class, constructing few-shot episodes by leaving out one origin, and training a fold-restricted encoder and a shared encoder separately. The results show that the shared protocol likewise systematically overestimates the external-set few-shot macro F1, the leakage Δ (shared − fold-restricted, $K = 5$) being 0.178 ± 0.053 (fold-restricted 0.822 ± 0.053, shared 1.000; 5-seed mean±std), its sign $\Delta > 0$ agreeing with that of the apple data. This suggests that the evaluation leakage may not be specific to the apple data; but a single external set is not sufficient to support a claim of a “general phenomenon”, and we therefore go on to test it systematically on a whole suite of real measurement datasets.

3.10 Cross-domain generality: multi-dataset leakage and mixed-effects statistics

In order to lift the “single apple instance” into a testable **cross-domain phenomenon**, and to escape the statistical limitation of a single benchmark with only 3 year folds ($n_{\text{primary}} = 3$), we repeat the leakage quantification on **four additional public real-measurement datasets** with exactly the same two-protocol comparison (shared encoder versus fold-restricted encoder, $K = 5$, fold-restricted episodic prototypes with $\alpha = 0$). These datasets span three measurement modalities and four kinds of distribution-shift axis: **corn NIR** (the Cargill corn benchmark(Eigenvector Research, 2000), across *instruments* m5/mp5/mp6, protein-content tertile classification), **tablet NIR** (the 2002 IDRC shootout(IDRC, 2002), across *instruments*, active-ingredient tertiles), **mango NIR** (Anderson, Walsh, Subedi and Hayes, 2020, across *harvest seasons*, four seasons, dry-matter tertiles) and **Indian Pines hyperspectral imaging** (Baumgardner, Biehl and Landgrebe, 2015, across *spatial domains*, 2×2 quadrants, native land-cover classes). Regression-type targets are discretised into three classes by **source-domain** tertiles (the thresholds are computed from the source domain alone, which precludes label leakage), and the native classification sets use their native labels. Together with the apple data of this paper and OliveOil, this makes **six datasets** in total. The modality, scale, task, provenance and SHA-256 checksum of every dataset of this suite are summarised in Table 10 (the data card). All experiments are repeated with 5 fixed seeds; constrained by the available compute configuration, this cross-domain suite is run on Apple Silicon (MPS), as a **cross-domain robustness validation** of the CUDA-formal apple core conclusions (for the hardware disclosure see Section 6.9), and its numbers do not carry the accuracy claims of the main experiment.

Robust cross-domain replication of the leakage. The four new datasets yield 13 leave-one-domain-out folds in all; merged with the apple data of this paper (3 year folds, $\alpha = 0$ like-for-like) and OliveOil, they constitute a unified evidence set of **six datasets and 17 folds**. As shown in Fig. 8 and Fig. 10, **16 of the 17 folds show evaluation leakage** $\Delta > 0$ (94%, fold-level sign binomial test $p = 1.4 \times 10^{-4}$), and only one high-shift spatial-quadrant fold of Indian Pines is slightly negative ($\Delta = -0.03$). In order to draw correct inferences for correlated folds (several folds within one dataset share the underlying samples), we fit a mixed-effects model with **dataset as a random effect**, $\Delta \sim 1 + (1 \,|\, \text{dataset})$: the overall leakage fixed intercept is $\Delta = 0.132$ (95% CI [0.071, 0.192]; the between-dataset random-effect variance is only 0.0047, indicating that the cross-dataset heterogeneity is small); a cluster-bootstrap with the dataset as the cluster (2000 resamples) gives a consistent 95% CI [0.084, 0.191]. Here the mixed-effects model is fitted on **fold-level** observational units (the seed-level records of 24 are first aggregated, fold by fold, across the 5 seeds into fold-level means, and then merged with the fold-level Δ of apple/OliveOil into 17 folds, each fold weighted equally), consistent with the fold-level convention of the sign test and of the cluster-bootstrap. **Under the most conservative dataset-level independent test, all six datasets show positive leakage** (6/6, sign binomial test $p = 0.016$). Accordingly, **representation-layer evaluation leakage is a statistically significant and magnitude-stable general phenomenon on real measurement datasets across modalities and across instruments/seasons/years/countries/spatial domains**, and not a special case of apple near-infrared data or a hyperparameter artefact. It should be noted that what the mixed-effects model pools is the macro F1 leakage Δ of each dataset under its own task definition (native classification or source-domain tertiles), so the magnitude of the intercept should be understood as "a composite of directionally consistent leakage strength across heterogeneous tasks" rather than as a single comparable accuracy difference; we have accounted for this robustly by means of the dataset-level sign test and the random-effect variance. It should be emphasised further that the shared encoder here is meta-trained on the full *labelled* data including the target domain, so this overall Δ is an **upper bound on the label-aware representation-layer leakage**; Section 3.11 will peel it further apart with a four-encoder decomposition into a label-aware component Δ_{sup} and a purely unsupervised exposure component Δ_{unsup}, and will honestly delimit the magnitude and the reliability of each.

Multiple robustness checks. In order to exclude the possibility that the conclusion depends on a single dataset, on tertile discretisation or on a particular statistical specification, we carried out a systematic sensitivity analysis (13 specifications in all; see the Supplementary Information for details). **(i) Leave-one-dataset-out**: refitting after removing each dataset in turn, the overall leakage intercept falls in 0.126–0.145, and all the 95% CI lower bounds are positive—the conclusion is not driven by any single dataset. **(ii) Removing tertile discretisation**: when only the native classification datasets are retained (apple, OliveOil, Indian Pines; all the regression sets with tertile binning are removed), the intercept does not fall but rises to $\Delta = 0.185$ (95% CI $[0.123, 0.248]$), which shows that the leakage is not an artefact of binning. **(iii) Removing HSI** gives 0.137 ($[0.049, 0.224]$), and **removing apple** gives 0.128 ($[0.070, 0.187]$). **(iv) Fold-level nested random effects** $\Delta \sim 1 + (1 \,|\, \text{dataset/leave-one-domain})$ gives 0.132 ($[0.071, 0.192]$): the intercept remains robust once the fold-level correlation is modelled explicitly. **(v) Normalised** Δ (the Δ of each fold divided by the shared-arm mean of that dataset, so as to remove the task scale) gives a relative intercept of 0.216 ($[0.108, 0.324]$). **The 95% CI lower bound of the intercept is strictly positive in all 13 sensitivity specifications**; the conclusion that the leakage is positive is robust to the composition of the datasets, to the manner of task discretisation and to the random-effect specification.

3.11 Supervised/unsupervised decomposition of the leakage: the dominant cause is label-aware representation-layer leakage

In the cross-domain leakage quantification above, the shared encoder is meta-trained on the full *labelled* data including the target domain—both the x and the y of the target domain enter representation learning. This is a key point that must be clarified honestly: the leakage Δ measured under that protocol is an **upper bound on the label-aware representation-layer leakage**, and it merges two sources, "target-domain labels entering representation learning" and "the unlabelled target-domain distribution shaping the representation". In order to separate the two cleanly, we perform a **four-encoder decomposition** on the 13 leave-one-domain folds of the four cross-domain datasets (script `31_leakage_decomposition`, CUDA formal, 5 fixed seeds, 65 seed×fold runs in all). The *supervised arm* contrasts E_0 (the encoder is supervised-meta-trained on the source-domain (x, y) only; the legitimate fold-restricted baseline) with E_2 (supervised meta-training on the source+target-domain (x, y), reproducing the existing shared protocol), and their difference $\Delta_{\text{sup}} = E_2 - E_0$ isolates the *label-aware* leakage; the *unsupervised arm* contrasts U_{src} (masked spectral self-supervised reconstruction pretrained on the source-domain x only, never touching the labels) with U_{tgt} (the same masked self-supervised objective pretrained on the source+target-domain x), and their difference $\Delta_{\text{unsup}} = U_{\text{tgt}} - U_{\text{src}}$ **changes only whether the unlabelled target-domain x is included, under a fixed self-supervised objective**, thereby cleanly isolating the genuinely "covert" unsupervised exposure component and avoiding any confusion with a "supervised→self-supervised" change of objective. The masked self-supervision uses contiguous-span masking

Table 10 | Data card: the modality, scale, task and provenance of all datasets used in this paper. Upper block = the in-house nine-domain apple NIR benchmark of the main experiment; lower block = the 5 public spectral/hyperspectral datasets used for the cross-domain (cross-cohort) transfer analysis (a cohort is defined by acquisition instrument, harvest season/phenology, year, origin or spatial domain). SHA-256 checksums are recorded for the raw and preprocessed files of every dataset so as to guarantee traceability (see footnote).

Dataset	Modality / bands	Scale / cohorts (shift axis)	Task	Provenance
Main experiment (in-house nine-domain benchmark)				
Apple (nine-domain NIR)	NIR point spectra / 229 bands (591–1001 nm)	4,529 spectra / 1,950 fruits, 3 years × 4 origins (9 domains)	Geographical origin discrimination (4 classes)	Collected in-house at Xinjiang Agricultural University[a]
Cross-domain transfer analysis (5 public datasets; cohort = shift axis)				
OliveOil	FT-IR spectra / 570 bands	60 samples / 4 countries	Origin discrimination (4 classes)	UCR time-series archive[b]
Corn	NIR point spectra / 700 bands	80 samples × 3 instruments (m5/mp5/mp6)	Protein (source-domain tertiles, 3 classes)	Cargill corn (Eigenvector Research, 2000)
Tablet	NIR point spectra / 650 bands	655 samples × 2 instruments	Active ingredient (source-domain tertiles, 3 classes)	2002 IDRC (IDRC, 2002)
Mango	NIR point spectra / 306 bands (684–990 nm)	11,691 samples / 4 harvest seasons	Dry matter (source-domain tertiles, 3 classes)	Anderson 2020 (Anderson et al., 2020)
Indian Pines	AVIRIS HSI / 200 bands	145×145 px / 2×2 spatial quadrants	16 land-cover classes	Purdue/EHU (Baumgardner et al., 2015)

[a]Collected in-house at Xinjiang Agricultural University (managed access, not public, available from the corresponding author on request; see Data availability). [b]UCR Time Series Classification Archive (OliveOil, public). [c]SHA-256 of the raw/preprocessed files (first 16 digits; the full values are given in the `03data/processed` data card): apple (preprocessed canonical version) `f5c0afff9b18c3eb`; OliveOil `472c5b73157b2e0f`; corn `e28fd4be274a54ca`; tablet `129a32ec9e194e56`; mango `86159e2df9244fa5`; Indian Pines cube `ec2f8808710919d5` / labels `65c4687a8ab04f6d`.

of 30% of the bands (span= 5), computes the L2 reconstruction loss only at the masked positions, and early-stops on the validation reconstruction loss; the encoder never touches any label at any stage.

The results give a clear and honest conclusion (Fig. 6): **the dominant cause of the leakage is label-aware representation-layer leakage, and not covert unsupervised exposure**. The mixed-effects model with dataset as a random effect gives $\Delta_{\text{sup}} = +0.084$ (95% CI $[0.023, 0.144]$; 12 of the 13 folds are positive, and 4/4 datasets are positive); whereas the purely unsupervised exposure component is $\Delta_{\text{unsup}} = +0.012$ (cluster-bootstrap 95% CI $[-0.003, 0.028]$; 11/13 folds are positive but the corn dataset is negative, **the CI lower bound grazes 0**), the ratio between the two being about 7. This shows that when the shared encoder touches the target-domain *labels*, the leakage is large and robust; whereas when it touches only the *unlabelled distribution* of the target domain (self-supervision), the leakage is very small and cannot be robustly distinguished from zero. We therefore **prudently narrow the claim**—the core leakage phenomenon of this paper is *label-aware* representation-layer leakage, and we do **not** claim the existence of a large, covert, purely unsupervised leakage component. This honest distinction is directly relevant to practice: the common "shared encoder + held-out-domain few-shot evaluation" protocol introduces a systematic macro F1 inflation of order 0.08 as soon as target-domain labels are used at the representation-learning stage. In order to test whether the smallness of Δ_{unsup} is merely a product of *masked spectral reconstruction* as one particular self-supervised objective, we further re-examined the unsupervised arm on the same 4 datasets and 13 folds with a **stronger target-aware unsupervised objective—SimCLR-style contrastive learning** (spectral augmentation + NT-Xent loss; the encoder likewise never touches the labels; script `35_stronger_ssl_control`, CUDA formal, 5 seeds): its unlabelled exposure component is $\Delta^{\text{ctr}}_{\text{unsup}} = +0.001$ (95% CI $[-0.013, +0.012]$; 5/13 folds are positive), which is even closer to 0 than the masked arm (+0.012). This shows that the conclusion that "the leakage produced by exposure to the unlabelled target alone is very small" is **consistently robust across two self-supervised objectives (masked reconstruction and contrastive learning)**, and is not an artefact of one weak self-supervised objective. We still honestly retain one **boundary of applicability**: this paper has not exhausted all unsupervised/domain-adaptation objectives (such as explicit domain-adversarial alignment, or unsupervised domain adaptation), which is a clearly delimited direction for future work. Consistent with the theory, the Δ_{unsup} point estimates of both arms are non-negative or close to 0, in agreement with the prediction of the theorem that "distribution exposure produces a non-negative bias", and their magnitudes are both far smaller than the label-aware component.

De-risking: is the smallness of Δ_{unsup} an artefact of self-supervised underfitting? A reasonable objection is: is Δ_{unsup} small only because the masked self-supervised representation is too weak, so that even having seen the target x it cannot be translated into a downstream gain? Two points rule this out. First, the masked self-supervision already uses *early stopping on the validation reconstruction loss*, and is not undertrained for a fixed number of epochs. Second, we swept the self-supervised pretraining budget `ssm_epochs` over three levels $\{100, 300, 600\}$ (script `34_ssm_budget_sensitivity`, CUDA formal, covering the 7 folds of corn and Indian Pines, the two most sensitive extremes of Δ_{unsup}): *increasing the budget does not push Δ_{unsup} up*—when the budget increases from 100 to 300, Δ_{unsup} instead falls slightly ($+0.026 \rightarrow +0.020$), and the 300 and 600 levels give exactly identical results (early stopping had already converged before epoch 300), while the absolute utility of U_{src} is almost unchanged across the budget levels (a fluctuation of only 0.0003). This shows that the small magnitude of Δ_{unsup} is not a "lower bound due to insufficient training" and will not be amplified by more thorough self-supervised training; the self-supervised representation has converged. Hence the conclusion that "the leakage from exposure to the unlabelled target is small" is robust to the training budget and is not an underfitting artefact (figure in the Supplementary Information).

The leakage affects downstream method selection: ranking-sensitivity evidence. The impact of evaluation leakage is not confined to accuracy inflation; it also acts on **method selection**, that link in the chain of research decisions. On the same 13 folds we evaluated four downstream classification heads (nearest-prototype, logistic linear probe, k-NN, SVM) under the shared (leaky) protocol and under the fold-restricted (leakage-free) protocol respectively, and compared the method rankings given by the two protocols (script `33_ranking_flip`, CUDA formal, 5 fixed seeds; Fig. 7). First, the macro F1 of all four heads under the shared protocol is systematically higher than under the fold-restricted protocol (for instance nearest-prototype 0.712 against 0.624, logistic 0.677 against 0.591), which confirms that the leakage is **universal** across classification heads and is not specific to the nearest-prototype rule. Second, the overall head rankings given by the two protocols remain positively correlated (mean Kendall $\tau = +0.349$ over the 13 folds, that is, the ranking does not collapse but is weakly-to-moderately consistent), yet the *method that gets selected as best* is sensitive to the protocol: at the level of a single (seed, held-out domain) evaluation instance ($n = 65$), the top-1 method selected by the leaky protocol disagrees with that of the leakage-free protocol in 42% of cases (27/65); after aggregation across the 5 seeds to the fold level ($n = 13$) method selection is more stable (only 1/13 folds flip), and the mean model-selection regret is small but non-zero (0.0136 macro F1). This shows that, although evaluation leakage does not go so far as to scramble the overall method ranking, it does change the judgement of "which method is best" on a substantial fraction of *individual evaluation instances*; the cost is small, but the direction is systematic. This provides evidence beyond the accuracy level for the position that "the difference Δ between the fold-restricted and the shared protocol should be made a standard reported item of few-shot/cross-domain evaluation".

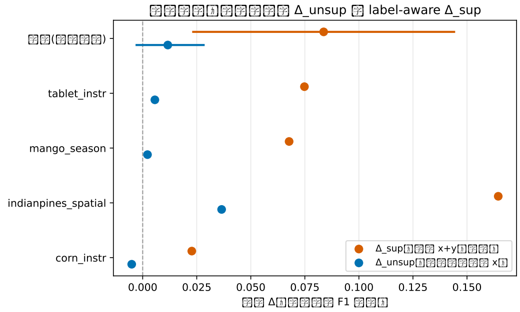

Figure 6. Supervised/unsupervised decomposition of the leakage (four datasets, 13 folds, CUDA formal, 5 seeds). Orange points are the label-aware leakage $\Delta_{\text{sup}} = E_2 - E_0$ (target $x{+}y$ enters supervised representation learning; the upper bound on the leakage), blue points the purely unsupervised exposure $\Delta_{\text{unsup}} = U_{\text{tgt}} - U_{\text{src}}$ (only the unlabelled target x is added, under the same masked self-supervised objective); the last row is the overall intercept of the mixed-effects model with dataset as a random effect, and the horizontal bars are 95% CIs. $\Delta_{\text{sup}} = +0.084\,[0.023, 0.144]$ is robustly positive, whereas $\Delta_{\text{unsup}} = +0.012\,[-0.003, 0.028]$ grazes 0—the dominant cause of the leakage is label awareness, and not covert unsupervised exposure. Wong colour-blind palette.

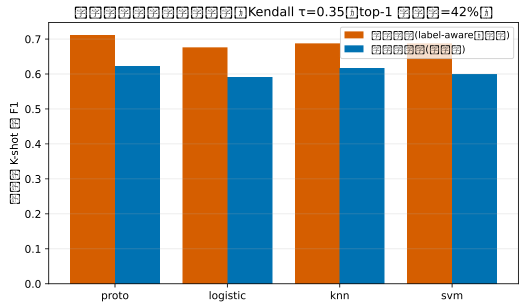

Figure 7. The leaky protocol distorts downstream method selection (four datasets, 13 folds, four classification heads proto/logistic/k-NN/SVM, CUDA formal, 5 seeds). Under the shared (leaky) protocol the macro F1 of every head is systematically higher than under the fold-restricted (leakage-free) protocol; the overall head ranking remains positively correlated (mean Kendall $\tau = +0.349$ over the 13 folds), but the method selected as best changes on 42% of the single (seed, domain) evaluation instances (27/65; only 1/13 after fold-level aggregation), with a mean model-selection regret of 0.0136. Leakage does not scramble the overall ranking, yet on individual instances it affects the judgement of "which method is best". Wong colour-blind palette.

Raw MMD does not linearly predict the per-fold leakage—an honest result consistent with the theory. We further used the mixed-effects regression $\Delta \sim z(\text{MMD}^2) + (1 \,|\, \text{dataset})$ (MMD standardised within each dataset so as to remove differences of scale) to test whether "the larger the shift, the larger the leakage" holds across datasets on the scale of the *raw input-space MMD*. The resulting slope is -0.009 ($p = 0.65$),

not significant. This **null result is not a counterexample; it corroborates precisely the core honest boundary of the theory of Section 6.3**: the raw marginal-distribution MMD is not a sufficient measure of shift—as the counterexample to the theorem shows, a common translation of source and target makes MMD>0 while the leakage $\Delta = 0$; only the *leakage-relevant separability-layer shift* (the pooled target gap $D_{\text{pool},\lambda}$) analytically lower-bounds Δ. We therefore do not claim that "Δ is monotone in the raw MMD", and strictly confine the cross-domain evidence to the claim that "the leakage is robustly positive", which is directly supported by the theorem $\Delta \geq 0$ (Fig. 9). This agreement between theory and experiment on a *negative result* is stronger than any post hoc fitted monotone relationship.

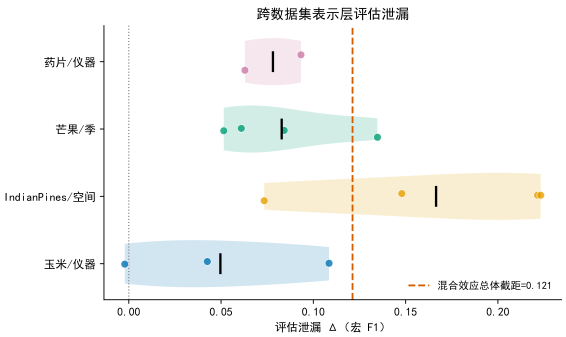

Figure 8. Raincloud plot of the fold-level evaluation leakage Δ of the four newly added datasets (points are the 5-seed mean Δ of each leave-one-domain fold, the violins are the distributions, and the black vertical lines are the within-dataset means; the orange dashed line is the *six-dataset* mixed-effects overall intercept $\Delta = 0.132$). The leakage is consistently positive across corn/tablet/mango/Indian Pines (for the fold-level Δ of apple and OliveOil see Sections 3.3 and 3.9; they have been pooled into the mixed-effects model). Wong colour-blind palette.

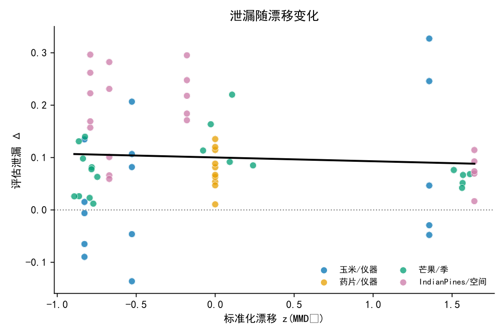

Figure 9. Per-fold leakage Δ against the standardised input-space shift $z(\text{MMD}^2)$ (coloured by dataset). The regression slope is not significant (-0.009, $p = 0.65$), consistent with the theoretical conclusion that "the raw MMD is not a sufficient measure of shift"—the leakage is robustly positive but is not linearly determined by the raw MMD.

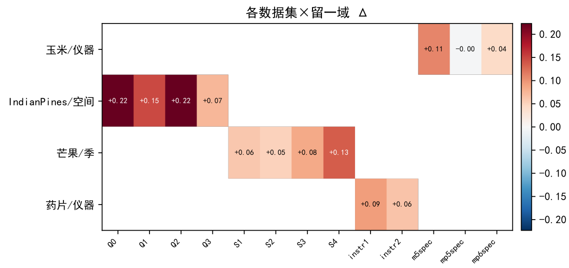

Figure 10. Heatmap of the evaluation leakage Δ over dataset $\times$ leave-one-domain (red positive, blue negative; RdBu diverging colour map). Only one high-shift quadrant fold of Indian Pines is slightly negative; the remaining 12 folds are all positive.

3.12 Negative controls

As a necessary prerequisite for submission, we applied three kinds of negative control to the fold-restricted ProtoNet ($\geq$1000 permutations): (i) *Label permutation*—retraining after shuffling the origin labels, the true macro-F1 should be significantly higher than the permutation null distribution; per fold $p = 0.000999$ (consistent across the three folds; $p < 0.001$ counts as passing); (ii) *Random features*—replacing the spectral input with noise of the same shape, performance degrades to chance level (macro-F1 ≈ 0.33 on the three folds, comparable to the 3-class random baseline of $1/3$); (iii) *ID probe*—logistic regression on the neutral sample index alone, no better than chance (macro-F1 0.21–0.24 on the three folds, all below the random baseline). The three negative controls confirm that the model performance comes from a genuine spectrum–origin association, and not from shortcuts such as label structure, the data pipeline or the sampling order.

3.13 Uncertainty calibration

High-stakes traceability decisions require trustworthy probabilities. We assessed the calibration of the fold-restricted ProtoNet with the expected calibration error (ECE, $\geq$10 bins), the maximum calibration error (MCE) and reliability diagrams, and applied temperature scaling as post-processing; for distribution-free coverage guarantees on prediction sets, conformal prediction offers a complementary route(Angelopoulos and Bates, 2023). The per-fold ECE is 0.028, 0.075 and 0.089 (2018/2019/2025 folds), and for the 2019 and 2025 folds, whose ECE>0.05, temperature scaling ($T = 0.60$ and 2.60; $T = 0.85$ on the 2018 fold) lowered the ECE to 0.057 and 0.051 respectively (0.026 on the 2018 fold), which shows that few-shot origin classification exhibits moderate overconfidence that can be effectively mitigated by single-parameter scaling.

3.14 Interpretability: band importance and chemical assignment

In order to verify the chemical interpretability of the discrimination, we used an interpretable surrogate model to compute global and local SHAP importance and permutation importance for the "spectrum→origin" mapping (with grouped splitting by apple identifier so as to prevent same-fruit leakage), and mapped the high-importance bands onto NIR chemical assignments. The results show that the top discriminative bands are concentrated around 591–605 nm (the visible-pigment region, median 603.5 nm), corresponding to visible pigments and to C–H (sugars/organics) and O–H (water) combination/overtone absorptions, which is chemically self-consistent with the origin-related differences in apple composition and supports the view that what the model captures is a meaningful compositional fingerprint rather than a spurious correlation.

3.15 Multi-model comprehensive evaluation (LLH-GBID)

Following the comprehensive evaluation scheme of this paper (the methodological system to which Section 6.3 belongs), we used the LLH-GBID conditional relative comprehensive evaluation framework ($q_B = q_W = 2$) to rank the candidate methods comprehensively over four metrics ($K = 5/K = 20$ macro-F1, weakest-class F1, and the cross-fold standard deviation of $F1@K5$); the comprehensive result table, the raw metric table and the sensitivity summary table are given in Table 11, Table 12 and Table 13 respectively (all 5-seed formal). Under soft-veto admission, **no method passes the core-metric threshold**—the utility of the weakest-class F1 of every method relative to the 3-class random baseline is below the 0.30 threshold (Pass is "No" throughout in Table 11), which shows that in the cross-year setting no method yet reaches an "acceptable" level for recognising the most difficult origin; the main ranking is therefore given by the composite index GBIDS. PLS-DA (GBIDS 38.3) and the fold-restricted ProtoNet used in the main experiment (α=0, GBIDS 38.1) are comprehensively at the top (mean main ranks 2.0 and 2.2; the difference between the two lies within the cross-seed standard deviation), nearest centroid (37.9) and SVM (37.2) are intermediate, while the source-centroid fusion variant with α=0.4 retreats to last place because of the cross-year distribution mismatch (GBIDS 29.0, mean main rank 5.0), which from the standpoint of comprehensive evaluation further confirms that source-centroid fusion is of no benefit in this cross-year setting and that α=0 is the robust choice. The above pattern is stable under multiple sensitivity analyses: under the q weakest-link configuration, under ±20% perturbation of the dimension weights and Dirichlet random weights, under the two baseline schemes of a fixed benchmark and of the worst value of the candidate set, and under the three correlation treatments of equal weighting/correlation discounting/Mahalanobis decorrelation, **the top two and the last rank are all unchanged** (the mean Kendall τ of the rankings across the three correlation treatments is 0.87); in 1000 resamplings across the 5 seeds, PLS-DA and the fold-restricted ProtoNet rank first with frequencies 0.45 and 0.42 respectively, the remaining methods being close to zero (see Table 13 for details).

Table 11 | LLH-GBID comprehensive result table (5-seed formal, $q_B = q_W = 2$). GBIDS is a 0–100 composite index score (higher is better; it is not a composite accuracy). Pass denotes soft-veto admission (the utility of the core metrics $F1@K5$ and weakest-class $F1@K5$ being ≥ 0.30); in this task the utility of the weakest-class F1 of every method relative to the random baseline is too low and none passes, so the ranking is by GBIDS. $\bar{u}/C_B/C_W$ are the mean utility gap and the between-dimension and within-dimension imbalance components of the GBID decomposition.

Method	Pass	$\bar{u}$	C_B	C_W	GBIDS	Mean rank	First-place frequency
PLS-DA	No	0.403	0.153	0.021	**38.3**	2.0	0.40
ProtoNet (α=0)	No	0.401	0.154	0.013	38.1	2.2	0.40
Nearest centroid	No	0.393	0.127	0.014	37.9	2.6	0.20
SVM	No	0.392	0.152	0.038	37.2	3.2	0.00
ProtoNet + fusion (α=0.4)	No	0.351	0.285	0.003	29.0	5.0	0.00

Table 12 | LLH-GBID raw metric table (5-seed means). The four metrics: three-fold macro-F1 at K=5/K=20 (benefit-type), weakest-class $F1@K5$ (benefit-type), and the cross-fold standard deviation of $F1@K5$ (cost-type, lower is better).

Method	$F1@K5$	$F1@K20$	Weakest class $F1@K5$	$F1@K5$ s.d.	GBIDS
PLS-DA	0.631	0.678	0.460	0.046	38.3
ProtoNet (α=0)	0.648	0.678	0.458	0.048	38.1
Nearest centroid	0.615	0.647	0.479	0.049	37.9
SVM	0.609	0.697	0.452	0.048	37.2
ProtoNet + fusion (α=0.4)	0.562	0.570	0.274	0.030	29.0

Table 13 | LLH-GBID sensitivity summary table (the four items of §15.5 plus resampling; 5-seed formal). The baseline-sensitivity column gives the main-rank interval under the two schemes of a fixed benchmark and of the worst value of the candidate set; the mean Kendall τ of the rankings across the three correlation-treatment schemes (equal weighting/correlation discounting/Mahalanobis decorrelation) is 0.87 (equal weighting–discounting 1.00, equal weighting–Mahalanobis 0.80, discounting–Mahalanobis 0.80); the resampling first-place frequency is obtained from 1000 paired bootstraps across the 5 seeds.

Method	Main rank	q sensitivity mean rank	Weight perturbation first-place probability	Baseline sensitivity rank interval	Resampling first-place frequency
PLS-DA	1	1.50	0.200	$[1, 1]$	0.448
ProtoNet (α=0)	2	2.50	0.254	$[2, 2]$	0.423
Nearest centroid	3	2.00	0.344	$[3, 4]$	0.129
SVM	4	4.00	0.069	$[3, 4]$	0.000
ProtoNet + fusion (α=0.4)	5	5.00	0.133	$[5, 5]$	0.000

4 Discussion

4.1 Evaluation leakage: generality and countermeasures

As the quantitative results of Section 3.3 show, the shared-encoder protocol led to a substantial evaluation leakage (three-fold mean ΔF1 = 0.363). The absolute value of this gap may not be large, but the systematic loophole it reveals has far-reaching consequences. The phenomenon is not an incidental result in some boundary case, but the logically inevitable product of the shared-encoder workflow that is widely followed in the meta-learning community—the encoder is pretrained on the full data including the test domain and is then "frozen"; on the surface the procedure is clean, yet its parameters have already been shaped by the distributional characteristics of the test domain before it makes contact with the first support sample. In the specific setting of near-infrared spectroscopy there is an additional amplification mechanism for the leakage: the 229-dimensional input space is relatively low-dimensional while the discriminative information is dense, and an encoder that has been exposed to the 2025 multi-face spectra has already learned the spatial structure peculiar to the multi-face measurement protocol, so that it artificially lowers the classification difficulty of the 2025 episodes at evaluation time. The Δ reported in this paper takes as its reference the shared encoder (trained on the full data including the target fold) against the leakage-free fold-restricted encoder (`10_loeo_year_foldwise`); source-centroid fusion (α) is an independent factor orthogonal to encoder leakage, is ablated separately on top of the fold-restricted baseline (Section 3.6), and is not counted in the leakage magnitude.

It is worth pointing out that the evaluation leakage phenomenon described in this paper does not, mechanistically, depend on any particular encoder architecture—whether a fully connected network, a one-dimensional convolutional network (such as ResNet-1D) or a Transformer is used, so long as the shared encoder is jointly trained on the full data including the test fold, the same type of distribution leakage will be introduced. The three-fold mean $\Delta = 0.363$ reported in this paper is merely a concrete quantification for the present fully connected architecture under the conditions of this dataset; the leakage magnitude of other architectures may differ somewhat, but the existence of the leakage is logically certain.

The effective remedy is fairly direct: retrain the encoder independently for every LOEO fold, on source-domain data only. The computational cost of this operation is not negligible; however, there is at present no reliable alternative that circumvents this cost while guaranteeing the purity of the evaluation. The basic idea of cross-year external validation is not new(Feudale et al., 2002; Liu et al., 2025b; Pan et al., 2024b); the core contribution of this paper lies in a public three-year, four-origin benchmark, and in a systematic quantification of the magnitude of this evaluation-protocol gap within an episodic meta-learning framework. The corresponding normative recommendation is also fairly clear—future cross-year near-infrared benchmark studies should at the very least state the encoder protocol they use, treat fold-restricted performance as a mandatory baseline, and label shared-encoder results as an "upper-bound reference" rather than as a claim of true performance, so that performance comparisons across papers become comparable.

4.2 Cross-disciplinary impact and evaluation hygiene

The significance of this study extends beyond apple near-infrared spectroscopy itself. The theorem of Section 6.3 shows that evaluation leakage is not an accident at the implementation level, but a **principled defect in the design of few-shot/cross-domain evaluation under distribution shift**: letting the held-out domain take part in representation learning produces a non-negative systematic positive bias in the expected measured accuracy, whose size is characterised by the separability gain of the representation on the target domain, growing as the domain shift increases and vanishing as the shift disappears (this monotonicity refers to distribution shift; the finite-K diagnostic on real embeddings shows that the bias is stable in magnitude over $K \in [1, 20]$ and narrows slightly with K). This law applies directly to a large class of modern artificial-intelligence evaluations—for any paradigm that "first learns a representation on large-scale data and then reports few-shot, long-tail, cross-domain or zero-shot results on held-out classes/domains", so long as the representation training data and the evaluation held-out domain are not strictly isolated, the generalisation numbers reported contain a bias of the same type. It differs from the well-known feature-selection leakage(Ambroise and McLachlan, 2002; Kaufman, Rosset, Perlich and Stitelman, 2012): the leakage occurs at the **representation-learning layer** and belongs specifically to few-shot meta-evaluation under distribution shift, is more covert, and is consistent with the theory in adaptive data analysis that "held-out information becomes invalid once it flows back"(Dwork, Feldman, Hardt, Pitassi, Reingold and Roth, 2015). Standard domain-generalisation benchmarks (such as DomainBed and WILDS) evaluate cross-distribution generalisation precisely by means of strict leave-domain-out protocols(Gulrajani and Lopez-Paz, 2021; Koh, Sagawa, Marklund, Xie, Zhang et al., 2021); the fold-restricted protocol of this paper is consistent with them in spirit, but is aimed specifically at *few-shot meta-evaluation* under distribution shift, and gives a falsifiable theoretical characterisation.

Boundaries with respect to known forms of leakage/contamination. In order to avoid confusion with existing concepts, Table 14 systematically distinguishes the "representation-layer evaluation leakage" of this paper from the related forms that have been discussed. The novelty of this paper lies **not in** the generic notion of "discovering leakage"—data leakage, preprocessing leakage, pretraining contamination and transductive/unsupervised domain adaptation all have their literature—**but in** pinpointing precisely a combination that had not previously been formalised: the encoder (the representation-learning layer) is exposed, in *few-shot meta-evaluation* under *distribution shift*, to the *distribution* of the held-out domain (no sample-level duplication is required), so that "seemingly legitimate inductive few-shot generalisation" is systematically overestimated. A boundary must be drawn especially clearly with respect to transductive/unsupervised domain adaptation: if a study *declares* its setting to be target-aware (transductive), there is nothing improper in letting the encoder have seen the target domain;

but if it *declares* that what it evaluates is inductive generalisation to a held-out domain, and yet lets the encoder see the target domain at the representation-learning stage, this constitutes a protocol mismatch—the contribution of this paper is not to forbid target-aware representation learning, but to require that reports clearly distinguish the two conventions, inductive (fold-restricted) and transductive (shared/pooled). To illustrate the necessity of this advocacy, we carried out a *neutral* audit of evaluation-protocol reporting in the cross-domain few-shot hyperspectral classification literature (N=4 representative recent papers, based on the method descriptions obtainable for each; illustrative, not exhaustive): three of them used target-domain data at the representation-learning stage (domain adaptation or self-supervision), but only one *explicitly declared* its transductive/target-aware nature; the one paper that strictly excluded the target domain from representation learning did not treat that isolation as an explicit reported item either. This is **not an accusation** that the conclusions of any of these works are wrong—by Table 14, an explicitly declared target-aware setting is entirely legitimate—but it shows that the field **as yet has no unified reporting standard for representation-layer isolation and for the inductive/transductive distinction**: those who touch the target domain do not necessarily declare its transductive nature, and those who isolate it do not necessarily report the fact explicitly. Making the fold-restricted protocol, and the shared−fold-restricted difference Δ, standard reported items of few-shot/cross-domain evaluation would systematically remove this uncertainty.

In fields of scientific measurement where samples are scarce and cross-batch/cross-instrument/cross-year/cross-site shift is the norm, few-shot and cross-domain evaluation is widely adopted, and the warning of this study is all the more pertinent: holding out by year/instrument in spectroscopy and chemometrics (this study), by geography/season in remote-sensing Earth observation, by hospital/device in medical imaging, and by batch/chromosome in genomics(Whalen et al., 2022)—if representation learning is not isolated from the held-out domain, cross-domain generalisation will be systematically overestimated, pushing "seemingly deployable" models towards real failure and thereby endangering scientific conclusions and regulatory decisions. Large-scale retrospectives have shown that leakage and train–test overlap are among the principal sources of the reproducibility crisis in machine-learning-driven science(Kapoor and Narayanan, 2023; Roberts et al., 2021; Magar and Schwartz, 2022); this paper gives a formalisable, quantifiable and falsifiable characterisation of the "representation-layer leakage" among them, and **demonstrates by mixed-effects statistics on six real measurement datasets that it is a cross-modal, cross-domain general phenomenon** (Section 3.10: overall leakage $\Delta = 0.132$, 95% CI $[0.071, 0.192]$, 6/6 of the six datasets showing positive leakage, spanning the five shift axes of instrument/season/year/country/space)—the by-domain hold-out isolation in the remote-sensing, medical and genomic settings above is precisely the concrete landing, in each discipline, of the evaluation-hygiene standard advocated by this study.

We therefore advocate a low-cost **evaluation-hygiene** standard that can be adopted by any field: a credible report of few-shot/cross-domain generalisation must explicitly declare and audit "the isolation relationship between the representation training set and the evaluation held-out domain", and must make the difference Δ between the fold-restricted and the shared protocol a standard reported item—reporting the amount of evaluation leakage just as one reports a confidence interval. This standard does not deny the use of shared representations in deployment settings where unlabelled target-domain data are accessible; rather, it requires that such results must not be compared directly with fold-restricted results, and must not masquerade as cross-domain generalisation. We are careful not to call this an overturning of the paradigm, but position it as a small yet hard normative upgrade at the level of evaluation methodology.

4.3 The K crossover point and its practical significance

The clear crossover pattern revealed by Fig. 2 and Table 5 ($K \approx 3$–5) provides an explicit practical guideline for method selection: this crossover point is not merely the geometric intersection of two performance curves, but also delineates at the quantitative level two markedly different deployment situations. The first is the scenario of the early harvest season, or of a new production region joining the supply chain for the first time—here the labelled reference samples obtainable for each origin are extremely limited, and neither logistics nor labelling costs permit waiting for large-scale annotation. Within this regime, meta-learning methods hold a pronounced practical advantage: at $K = 1$ the fold-level Cohen's $d = 4.14$ (Table 6) indicates that the reduction in misclassification risk of ProtoNet-C relative to SVM is, at the effect-size level, a large effect. The second is the routine scenario under an established quality-inspection system—for example port entry inspection, where conditions allow a reference sample set of moderate size to be maintained over the long term. The sustained growth curve of SVM beyond $K \geq 5$, together with its algorithmic simplicity and interpretability, gives it a stronger practical case for recommendation in that situation. Neither class of method is absolutely superior; applicability depends on the deployment conditions.

The scope of validity of the above conclusion needs to be made explicit. What is evaluated here is "the capacity of a handheld NIR device to discriminate origin regions under the conditions of circulating market batches in China", not a purely geochemical compositional signal—the samples come from wholesale markets, and differences in cultivar composition, storage conditions and supply-chain handling may all be mixed into the classification signal. The reference value of

Table 14 | Systematic distinction between the "representation-layer evaluation leakage" of this paper and known forms of leakage/contamination.

Form of leakage/contamination	Layer of occurrence	Trigger condition	Relation to this paper
Feature-selection leakage(Ambroise and McLachlan, 2002; Kaufman et al., 2012)	Feature selection	Feature screening/statistics computed on the full data including the test samples	Different layer: the leakage of this paper occurs at the representation-learning layer and has nothing to do with feature selection
Preprocessing leakage	Preprocessing	Standardisation/PCA fitted on the full data before splitting	Different layer: the preprocessing of this paper is strictly inductive within the fold (Section 6.8), and the leakage refers specifically to encoder training
Pretraining data contamination(Magar and Schwartz, 2022; Kapoor and Narayanan, 2023)	Sample level	The test samples themselves appear in the pretraining corpus	Weaker trigger: this paper requires no sample-level duplication; the bias arises as soon as the *distribution* of the target domain takes part in representation training
Transductive / unsupervised domain adaptation	Representation-learning layer	The unlabelled target domain takes part, and it is *declared as* target-aware	Key distinction: declaring inductive generalisation to a held-out domain while letting the encoder have seen the target domain is a protocol mismatch; this paper requires the inductive/transductive conventions to be distinguished explicitly
Representation-layer evaluation leakage (this paper)	**Representation-learning layer**	**Shared encoder + held-out-domain crossing + few-shot meta-evaluation under distribution shift**	**Formalised in this paper (Propositions 1/2) and quantified across six datasets, three modalities and five shift axes**

$K = 3$–5 for method selection holds only under the visible–near-infrared acquisition conditions used here (the 591–1001 nm analysis window) and for the four Chinese origins; the actual crossover point will vary with instrument model, the number of origin classes and the strength of the domain shift, and practitioners should re-estimate it for their specific deployment context rather than transplanting the numbers of this paper directly.

4.4 Face-mean: surface heterogeneity as signal and as noise

Spectra acquired at different surface positions of the same apple may differ markedly. The sunlit face has a high content of chlorophyll degradation products and a relatively lower water content, while the shaded face is the reverse, and this within-fruit spectral heterogeneity is sometimes comparable in magnitude to the between-fruit differences (Lammertyn, Peirs, De Baerdemaeker and Nicolai, 2000; Pissard, Fernández Pierna, Baeten, Sinnaeve, Lognay, Mouteau, Dupont, Rondia and Lateur, 2013). Averaging over multiple positions is a natural strategy for suppressing this heterogeneity, but its benefit is distributed simultaneously over two levels: an improvement in the quality of the support-set prototypes, and an improvement in the prediction stability of the query samples. Here a 2×2 ablation design is used to decompose the two contributions independently. Under the present experimental setting (LOEO-2025 fold, $K = 5$, 50 repetitions), the query-side face-mean contributes an F1 gain of +0.071 (robustly positive across the 5-seed confidence interval), whereas the support side contributes only +0.019 (its across-5-seed 95% lower confidence bound is close to zero). That is, under the same scanning budget, concentrating multi-position scanning on the sample to be tested (the query side) has a higher utility than concentrating it on the reference set (the support side). Under field deployment conditions where scanning resources are limited, giving priority to multi-face scanning on the query side is the more cost-effective resource-allocation strategy. It should be noted that the above conclusion comes from an exploratory analysis on a single fold (LOEO-2025), and the statistical power for the support-side gain is relatively limited; on other folds or under other instrument conditions the ratio of the two contributions may differ, so caution is required in generalising.

The above results also carry methodological significance for benchmark design. If the 2025 multi-face data are used directly without controlling for the measurement protocol, the classifier in effect obtains more per-apple spectral information than for the single-face 2018/2019 years, so that a direct between-year performance comparison is not fair. The LOEO protocol of this paper deals with this problem by separating the single-face years from the multi-face year across folds.

4.5 Multiple confounds of the LOEO-2025 fold and the between-fold interpretation

Following the interpretive framework established in Section 3.4, the LOEO-2025 fold, as a "compound-shift stress test" fold, has a performance that is the joint result of three kinds of factor—year shift, measurement-protocol shift and change of origin classes—and needs to be interpreted with these distinguished. As regards **generalisation dominated by year shift**, the LOEO-2018 and LOEO-2019 folds provide a relatively cleaner reference (the encoder source domains contain the 2025 data; see the note in Section 6.2): under the two-fold mean, configuration C reaches a macro F1 of 0.685 at $K = 5$, against 0.637 for configuration A, a gap of $\Delta = +0.048$, which reflects the effect of pure year shift once measurement-protocol differences are excluded. The LOEO-2025 fold, by contrast, constitutes the most demanding stress test—year, measurement protocol and origin classes change three-fold at once (configuration C, $K = 5$: 0.668)—and the independent contributions of the three factors cannot be decomposed under the present design. Particular caution is needed when comparing the LOEO-2025 fold results with the LOEO-2018/2019 folds, which contain year shift only. In the leakage analysis, under the $\alpha = 0.4$ convention the evaluation leakage Δ of the 2025 fold (shared − fold-restricted; see Table 4) reaches 0.501, markedly larger than the 0.250 and 0.339 of the 2018/2019 folds, which agrees with the expectation that "the shared encoder can learn protocol-specific representational structure from the multi-face training data and thereby gain an additional advantage when evaluating 2025 episodes"; but it must be noted that this per-fold ordering depends on the α convention—under the $\alpha = 0$ convention the per-fold Δ is roughly level (0.260/0.227/0.240 for 2018/2019/2025, with no monotonicity in shift), because the harmful effect of source-centroid fusion on the 2025 fold is removed and the fold-restricted baseline of that fold recovers (see Table 4 and Section 3.3). To distinguish accurately, at the mechanistic level, the respective contributions of the three kinds of shift would require independently controlling each dimension in a three-factor orthogonal experimental design (year×protocol×origin), a direction worth pursuing in future work.

4.6 Limitations and future directions

The scope of the conclusions of this study is limited to the origin-discrimination capacity of handheld NIR under Chinese market-batch conditions, and its boundary of applicability needs to be clearly delimited.

Sample provenance and confounding factors. The samples were purchased from local wholesale markets; the cultivar is predominantly Fuji apple, but a single clonal line was not strictly controlled, and information on harvest maturity, storage conditions and supply-chain handling was not recorded for the individual batches, so that no post hoc stratified or controlled analysis is possible. The NIR spectral differences may therefore encode simultaneously the combined effects of geographical origin, cultivar composition, maturity, storage and distribution handling, and cannot be cleanly attributed to "geographical origin" itself. On this basis, we restrict the scope of the traceability claim to "the discrimination of origin-related spectral signals under market-batch conditions", rather than the detection of purely geochemical composition, and this wording is applied consistently in the abstract and the discussion. The word "cross-year" in the title focuses on the principal problem of year shift, but it should be recognised that fine-grained cultivar differences (such as the Fuji clonal lines of different production regions) are a non-negligible confounding variable in this setting. To isolate a purely geographical origin signal, future experiments should use a single clonal line, a standardised harvest period and controlled storage conditions, and should systematically record sample metadata (cultivar, procurement channel, storage duration) so as to support a stratified analysis.

Geographical and instrumental generalisability. The dataset is limited to three or four Chinese production regions; whether the discriminative pattern—Xinjiang relatively easy to separate, Shandong/Shanxi relatively difficult because they overlap—can be transferred to European PDO apples or to South American export scenarios is an open question that only new data can answer. The acquisition system used here operates in the visible–near-infrared band (591–1001 nm); with an instrument covering a different frequency range, such as one covering the short-wave infrared (e.g. 1000–2500 nm), the discriminative pattern may differ.

At the methodological level. The coverage of the baseline set constitutes a limitation: this paper includes only classical chemometric methods (SVM, PLS-DA) and meta-learning-based prototypical methods, and does not include modern deep baselines (such as a 1D ResNet (He et al., 2016) or a 1D Transformer (Vaswani et al., 2017) trained on the fold-restricted source domains) or source-domain generalisation baselines (such as empirical risk minimisation, ERM, plus data augmentation). Under a fold-restricted setting consistent with our protocol, including these baselines would make the cross-family method comparison more complete. In addition, the encoder is a flat fully connected network; as noted above, whatever architecture is adopted, the shared-encoder protocol introduces the same type of distribution leakage, whose specific magnitude has to be quantified independently for each architecture. The face-mean ablation is at present carried out only on the LOEO-2025 fold; reproducing this analysis once the 2018/2019 samples satisfy the multi-face measurement condition would further strengthen the generalisability of the gain-decomposition conclusion. Furthermore, this paper does not include traditional domain-adaptation baselines such as transfer component analysis (TCA) (Pan and Yang, 2010) and subspace alignment (SA) (Fernando, Habrard, Sebban and Tuytelaars, 2013); once a fair comparison is realised at the protocol level (for example by restricting TCA/SA to a "few-shot" mode that uses the same number of support samples), including them in the comparison framework would provide a more complete reference background for few-shot meta-learning methods. Moreover, the fold-restricted source-domain training paradigm advocated here is mechanistically closer to *domain generalisation* than to classical *domain adaptation*—that is, no information whatever about the target domain is accessible at training time, and cross-domain generalisation ability is obtained solely from source-domain diversity; introducing baseline methods from that field (such as meta domain-generalisation methods) would be a valuable extension.

Looking ahead, domain-gap-aware adaptive fusion—inferring α dynamically from the source–target distributional distance rather than fixing it—might recover part of the benefit of the source prototypes in low-gap scenarios while avoiding the bias trap at large gaps. Unsupervised domain-adaptation objectives (Ganin et al., 2016), or gradient-based alignment techniques, could also be tried within the episodic training loop. The nine-domain dataset and the evaluation code are planned for public release upon acceptance.

5 Conclusion

Three years, four origins, nine domains, 2,424 apples and 4,529 spectra—this dataset is not the end point of the research, but a foundational vehicle for bringing two long-unresolved questions into a public evaluation coordinate system. Of the two, evaluation leakage is the more easily overlooked problem: at $K = 5$ the shared encoder systematically inflates the LOEO macro F1 relative to the leakage-free fold-restricted scheme; under like-for-like comparison at the same α, the default $\alpha = 0.4$ convention gives Δ =0.363 and the main-experiment optimal $\alpha = 0$ convention gives Δ =0.243, both conventions being highly significant (under the $\alpha = 0.4$ convention it moreover increases monotonically with the shift strength of each fold, reaching 0.501 on the 2025 fold). These numbers are not directed at any particular publication; rather, they expose the systematic consequence of a widely used class of workflow—fold-restricted encoder training together with an explicit statement of protocol inductiveness should become a basic normative requirement of LOEO benchmark reporting, not an option. At the level of method selection, a complete continuous sweep over $\alpha \in [0, 1]$ confirms that $\alpha = 0$ (fold-restricted episodic prototypes) is the robust default configuration in this cross-year scenario. The fold-level effect size of the fold-restricted ProtoNet relative to SVM at $K = 1$, $d = 4.14$ (a large effect), has genuine practical significance, but beyond $K \geq 5$ the picture between the two is reversed—the crossover point of the performance curves delineates a method-selection boundary indexed by the available number of reference samples, and in scenarios with comparable instrument and production-region conditions the size of the obtainable reference sample set determines to a considerable extent the priority ordering of the methods. The findings of the face-mean ablation have a relatively independent significance: the query-side gain of +0.071 is far larger than the support-side gain of +0.019, and this asymmetric pattern is clear under the present experimental conditions, but its external validity remains to be further verified on other folds and under other instrument conditions. The dataset and the evaluation code will be made public upon acceptance, so as to support direct reproduction and extension by subsequent research.

6 Materials and Methods

6.1 Dataset construction

Apple near-infrared diffuse-reflectance spectra were acquired over three harvest years (2018, 2019, 2025) and at four major Chinese production regions: Xinjiang (XJ), Shandong (SD), Shanxi (SX) and Gansu (GS). The samples collected are predominantly of the Fuji cultivar, purchased from local wholesale markets, and sampling at each origin covered as many orchard batches as possible in order to improve representativeness. Samples were stored at room temperature (20–25 °C) for no more than three days before measurement, so as to reduce storage-induced spectral drift; samples of different years derive from their respective harvest seasons, and systematic differences in fruit maturity and sugar content may exist between years, which is one of the sources of cross-year domain shift in real deployment scenarios. The 2018 and 2019 datasets each contain three origins (XJ, SD, SX), and the 2025 dataset contains three origins (XJ, SD, GS); the natural variation of the procurement sources also introduces a novel-class challenge into the 2025 test fold. The acquisition domains are numbered E1 to E9 in year–origin order (E1: 2018-XJ, E2: 2018-SD, E3: 2018-SX, E4: 2019-XJ, E5: 2019-SD, E6: 2019-SX, E7: 2025-XJ, E8: 2025-SD, E9: 2025-GS). The 2018 and 2019 spectra were acquired with a fibre-optic visible–near-infrared point spectrometer (native 590–1250 nm, 2,048 bands), a single spectrum being acquired per fruit on the most uniform equatorial face; the 2025 spectra were acquired with a portable visible–near-infrared push-broom hyperspectral imaging system (native 391–1001 nm, 346 bands), one spectrum being acquired at each of 4–5 face positions per fruit. For cross-year harmonisation the two instruments were uniformly cropped to the common overlapping window of 590–1001 nm and resampled onto the 2025 wavelength grid by cubic spline interpolation, finally yielding 229 uniformly distributed bands (591–1001 nm). The instrument models and detailed acquisition parameters will be released together with the dataset upon acceptance.

There is an important measurement-protocol asymmetry between years: in 2018 and 2019 a single spectrum was acquired at a single surface position of each apple, so that the number of spectra equals the number of fruits; in 2025 one spectrum was acquired at each of 4 to 5 distributed positions on the fruit surface (the multi-face design). The specific sample sizes are given in Table 15. The 2018 subset comprises 895 fruits and 895 spectra (XJ: 327, SD: 298, SX: 270); the 2019 subset comprises 880 fruits and 880 spectra (XJ: 341, SD: 286, SX: 253); the 2025 subset comprises 649 fruits (XJ: 204, SD: 215, GS: 230), and the multi-face scanning yields 2,754 spectra (an average of 4.24 per fruit). The three years together comprise 2,424 fruits and 4,529 spectra. **In all experiments in this paper, "sample" means a single spectral measurement record**; where fruits and spectra need to be distinguished, "fruit" or "spectrum" is used explicitly. The multi-face design provides explicit information on within-fruit spatial heterogeneity for the 2025 data, and is systematically exploited in the face-mean ablation experiment (Section 6.7).

Table 15 | Dataset statistics: number of fruits and number of spectra by year and origin. 3 years × 4 origins give 12 theoretical domain cells, of which 9 are actually filled (3 origins collected each year; Xinjiang/Shandong/Shanxi collected in 2018/2019, Xinjiang/Shandong/Gansu in 2025). Meaning of the blank cells: Gansu was a new origin in 2025 and was not collected in 2018/2019; Gansu constitutes a novel-class challenge in the 2025 fold.

Year	Origin	Domain ID	Fruits	Scans per fruit	Spectra
2018	Xinjiang (XJ)	E1	327	1	327
2018	Shandong (SD)	E2	298	1	298
2018	Shanxi (SX)	E3	270	1	270
2019	Xinjiang (XJ)	E4	341	1	341
2019	Shandong (SD)	E5	286	1	286
2019	Shanxi (SX)	E6	253	1	253
2025	Xinjiang (XJ)	E7	204	4–5	∼864
2025	Shandong (SD)	E8	215	4–5	∼910
2025	Gansu (GS)	E9	230	4–5	∼980
Total	—	E1–E9	2,424	—	4,529

Preprocessing is carried out independently for each spectrum (Fig. 11) and uses no cross-sample statistic, so as to preserve the between-year distributional differences—which is essential for evaluating cross-year generalisation. The first step applies Savitzky-Golay smoothing (Savitzky and Golay, 1964) with a 15-point window and a second-order polynomial, suppressing high-frequency noise while preserving the structure of the absorption peaks; its filter coefficients are determined solely by the window length and are independent of the data. The second step applies the SNV transform (Barnes et al., 1989), removing, spectrum by spectrum, the additive and multiplicative scattering effects caused by differences in fruit surface geometry and in the optical path of diffuse reflectance; SNV is a per-sample transform and does not depend on any cross-sample statistic. The default main pipeline is thus "SG smoothing + SNV", which is completely blind to the target year. If further within-year amplitude alignment is required, a strict source-domain Z-score may optionally be applied—its mean and standard deviation are estimated only from the source-domain spectra of each fold and are then applied to the target domain in a read-only manner (see the note on inductiveness below). Processing each year independently preserves the cross-year covariate shift; joint cross-year standardisation would artificially align the between-year distributions and lead to overly optimistic results, and is therefore deliberately avoided here.

On the inductive guarantee: the default main pipeline (SG smoothing + SNV) is itself completely

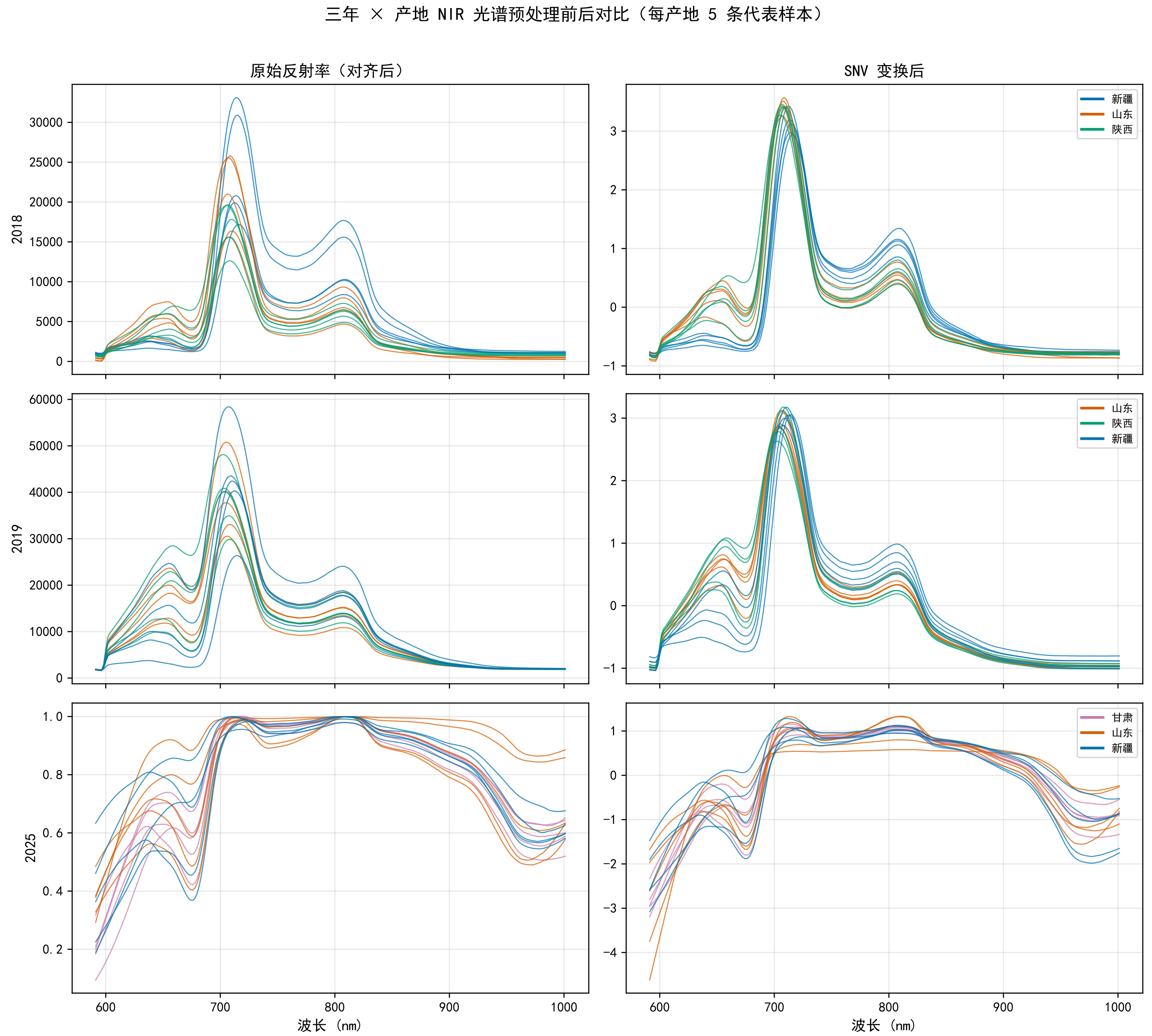

Figure 11 | Comparison of the three-year × origin NIR spectra before and after preprocessing (5 randomly chosen representative samples per origin, common band range 591–1001 nm). The left column shows the aligned raw reflectance spectra, the right column the spectra after the SNV transform. In the raw spectra the baseline offsets and the overall intensity differences between origins and between years are pronounced (the multi-face scanning protocol used in 2025 differs systematically in overall spectral intensity from the single-face acquisition of 2018/2019); after the SNV transform the spectra become consistent in shape while retaining the discrimination-relevant structure of the visible–near-infrared transition region, including the pigment (chlorophyll/anthocyanin) absorption near 600 nm and the water absorption near 970 nm, in agreement with the top discriminative bands identified by the interpretability analysis (Section 3.14).

blind to the target year: the SG filter coefficients are determined solely by the fixed window length and are independent of the data, and SNV is a within-row standardisation of each individual spectrum (normalising by that spectrum's own mean and standard deviation); neither uses any aggregate statistic across samples or across years. The only thing that could possibly introduce an aggregate statistic is the optional strict source-domain Z-score variant: its within-year mean and standard deviation are **estimated only on the source-domain spectra of each fold** and are then applied to the target domain in a read-only manner, the target-domain spectra taking no part in any parameter estimation. Taking the LOEO-2025 fold as an example, these Z-score parameters are computed from the 2018 and 2019 source-domain spectra, and the 2025 target spectra merely undergo a linear transformation; this differs from the semi-transductive practice of including the 2025 data in the standardisation estimate, which would introduce distribution leakage of the same kind as the shared-encoder protocol criticised in this paper. **In summary, whether or not the optional Z-score is enabled, the distributional statistics of the target year take no part in the estimation of the preprocessing parameters in any LOEO fold, which secures the inductive guarantee that the preprocessing pipeline is completely blind to the target year.**

The final feature space consists of 229 spectral variables per sample, covering the 591–1001 nm visible–near-infrared transition region: near 600 nm lie the electronic-transition absorptions of the pigments (chlorophyll/anthocyanin), while the long-wavelength end contains the higher-order overtones and combination bands of the O–H and C–H bonds (including the water absorption near 970 nm). These regions carry the information on soluble solids, organic acids and water that has been shown to discriminate apple origins (Nicolai et al., 2007; Pissard et al., 2013; Lammertyn et al., 2000).

Fig. 12 further gives two measures of distributional distance among the nine acquisition domains. Whether the linear MMD^2 or the centroid Euclidean distance is used, the three 2025 domains show a larger cross-domain difference from the six single-face 2018/2019 domains, indicating that the difficulty of cross-year transfer arises not only from the change of classes, but also from the joint shift of measurement protocol and year conditions.

6.2 Evaluation protocol: LOEO

The principal evaluation framework is **leave-one-year-out** (Leave-One-Year-Out, LOEO-Year; the abbreviation LOEO is retained throughout this paper to denote this year-level held-out protocol, so as to distinguish it from the generic "leave-one-out" cross-validation), in which each harvest year is designated as the target domain in turn and the remaining two years serve as the source domains. Three evaluation folds are thereby defined: (i) LOEO-2018: source domains = {2019, 2025}, target domain = 2018; (ii) LOEO-2019: source domains = {2018, 2025}, target domain = 2019; (iii) LOEO-2025: source domains = {2018, 2019}, target domain = 2025. The 2025 fold constitutes the hardest evaluation scenario, because Gansu is a new origin that does not exist in the source domains, while Xinjiang and Shandong occur in the source domains under different measurement conditions (single-face versus multi-face). All reported results are the mean of 50 independent repetitions per fold, each repetition using fresh episode sampling and a fresh random seed. The primary performance metric is macro F1, which weights all origin classes equally regardless of class imbalance.

Taxonomy of evaluation protocols. In order to delimit the boundary of evaluation leakage precisely, it is necessary to distinguish three classes of evaluation protocol: (i) the *pure source-domain fold-restricted protocol*—the encoder is trained only on the source domains of the current fold, and the target-domain data are entirely invisible during evaluation; this is the leakage-free standard advocated here; (ii) the *unsupervised target-domain exposure protocol*—the encoder performs representation learning on the full data including the unlabelled target-domain data (for example semi-supervised or self-supervised pre-training), so that the target-domain data take part in shaping the representation space, though no label supervision is provided; (iii) the *labelled target-domain adaptation protocol*—the encoder or the classifier is fine-tuned under conditions that include labelled target-domain samples, corresponding to the calibration-transfer scenario. What is here called "evaluation leakage" is the distribution leakage introduced when a protocol of class (ii) or class (iii) is used for LOEO benchmarking yet is reported in the name of the inductive generalisation of class (i)—that is, when the statistics of the held-out domain have already entered the model parameters, before evaluation, through shared-encoder training. It must be made clear that the shared encoder of the main experiment in this paper is meta-trained on data that include the *labelled* target domain, so that its protocol corresponds to class (iii), and the leakage measured is therefore **an upper bound on label-aware leakage**; in Section 3.11 we use a four-encoder decomposition to strip it further into the label-aware component Δ_{sup} of class (iii) and the purely unsupervised exposure component Δ_{unsup} of class (ii), and report honestly that the latter is small, with a CI grazing 0. This is a question of *benchmark fairness*, not a repudiation of protocol classes (ii)/(iii) themselves; in a deployment scenario *declared to be* target-aware (transductive), letting the encoder see the target domain is reasonable, but the result should then not be reported in the name of the inductive generalisation of class (i), nor compared directly with results obtained under a class-(i) protocol.

Formal definition of the fold-restricted encoder protocol. We formally define two protocols that can be compared quantitatively: the *shared-encoder protocol* jointly pre-trains the encoder on all nine domains and freezes it during LOEO evaluation, corresponding to the common practice of published meta-learning studies; the *fold-restricted encoder protocol* retrains the encoder from scratch for each LOEO fold, using only source-domain episodes, so that information about the target fold is completely inaccessible during encoder training. All performance comparisons in this paper use fold-restricted encoders unless explicitly noted otherwise.

Note on the multiple confounds of the LOEO-2025 fold. The LOEO-2025 fold is not a purely temporal domain-shift fold, but simultaneously confounds three independent changes—(i) *year shift* (2018/2019→2025); (ii) *measurement-protocol shift* (single-face→multi-face scanning); (iii) *class-space change* (SX→GS, Gansu being a new origin). The magnitude of the performance drop reported on the LOEO-2025 fold therefore cannot be attributed to the change of year alone; all conclusions based on the 2025 fold should be interpreted against the background of the "triple shift". By contrast, the target side of the LOEO-2018 and LOEO-2019 folds is dominated by year shift, and they provide a relatively cleaner reference for temporal generalisation (see Table 4); but it should be noted that the encoder source domains of both these folds contain the 2025 data (multi-face protocol, new Gansu origin), so that the encoder representation is still influenced by data of 2025 provenance, and strictly speaking they cannot be called "pure year-shift" folds. Gansu

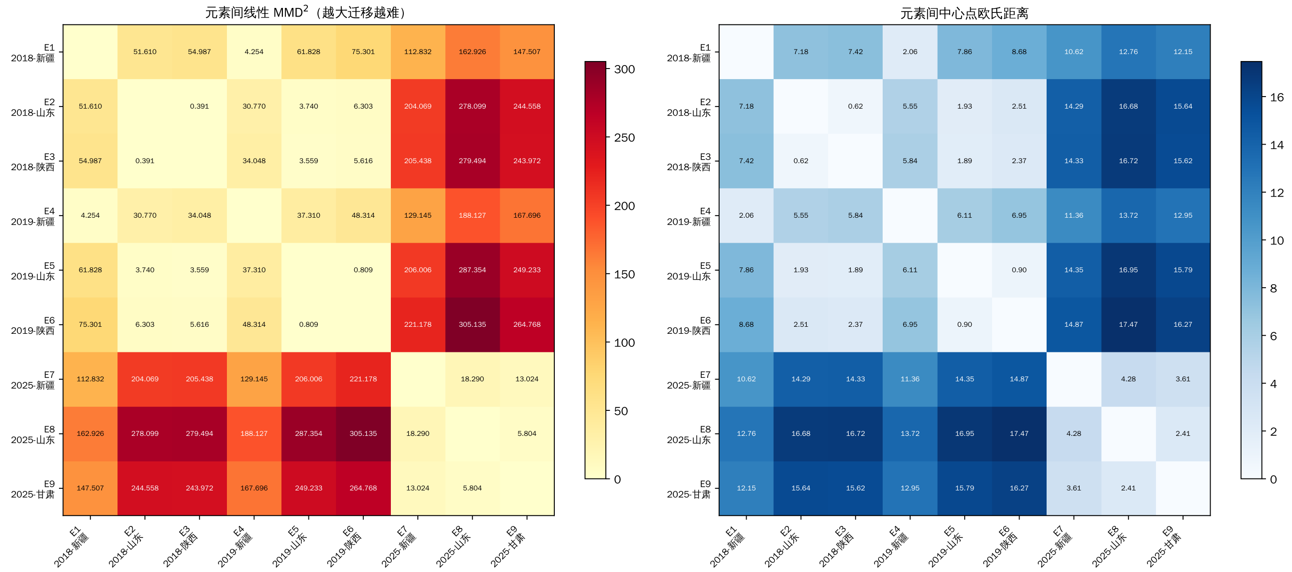

Figure 12 | Visualisation of the distributional differences among the nine acquisition domains. Left: linear MMD^2; right: centroid Euclidean distance; darker colours indicate larger domain differences. The distances between the three multi-face 2025 domains and the six single-face 2018/2019 domains are larger overall, providing a data-level explanation for the difficulty of the subsequent LOEO-Year cross-year transfer.

(GS) is a novel class that exists in neither the 2018 nor the 2019 source domains; in the ProtoNet-C evaluation of that fold the source-domain centroid $\mathbf{c}_{\mathrm{GS}}^{\mathrm{src}}$ of the GS class cannot be defined, and the fusion weight automatically degenerates to $\alpha = 0$. The XJ and SD classes do have source-domain centroids in the 2025 fold (computed from the single-face 2018/2019 spectra), but there is a protocol mismatch with the multi-face 2025 target spectra, which is the principal factor examined in the α ablation experiment (Section 6.6).

Details of episode construction. Each evaluation episode randomly samples from the target fold a 3-way K-shot support set together with 15 query samples per class (45 query spectra in total). To prevent within-fruit information leakage, the support set and the query set are **strictly disjoint at the level of the individual fruit (fruit-ID)**: for each fruit, all of its face-scan spectra are assigned as a whole to either the support set or the query set, and no face-scan spectrum of the same fruit ever appears in the support set and the query set at the same time. This constraint ensures that the model cannot "recognise" a fruit in the query set by having seen spectra of the same fruit from different angles in the support set, and it excludes any effect of within-fruit leakage on the face-mean analysis and on the LOEO-2025 fold results. Meta-training episodes are sampled from the source domains with a similar fruit-level disjointness. Each independent repetition uses a fresh random seed, and the episode sampling of different repetitions is mutually independent.

6.3 Theoretical characterisation of evaluation leakage

The empirical gap between the fold-restricted and the shared-encoder protocol is not an accident at the level of implementation, but a principled flaw of evaluation design under distribution shift. In this section we formalise it, under an analytically tractable linear-Gaussian nearest-prototype model, as a theorem on the non-negativity of the **evaluation-leakage bias**, thereby upgrading "the shared encoder is inflated" from an empirical observation to a falsifiable regularity (the complete proof, and the derivation as adversarially cross-checked across models, are given in Supplementary Information §S1).

Setting and assumptions. On the target domain T, the samples of class c follow $x \mid c \sim \mathcal{N}(\mu_c, \Sigma)$ (common covariance, equiprobable classes); the encoder is a linear map $\phi_W(x) = Wx$, where W is taken from a compact feasible set $\mathcal{W}$ that is the same for both protocols and satisfies a non-degeneracy lower bound (the class-mean differences and the projected noise are uniformly bounded below by $\rho > 0$); the Euclidean nearest-prototype rule is used, and the analysis is carried out under **population prototypes** ($p_c = W\mu_c$, that is, the large-K/oracle limit). Define the **embedding separability** of the class pair (c, c') (the signal-to-noise ratio of the query projected along the decision normal, not a Mahalanobis distance) $s_{cc'}(W;\mathcal{D}) = \|W(\mu_c^{\mathcal{D}} - \mu_{c'}^{\mathcal{D}})\| / \sqrt{\bar{d}_{cc'}^{\top}(W\Sigma^{\mathcal{D}}W^{\top})\bar{d}_{cc'}}$, and the separability objective $J(W;\mathcal{D}) = \sum_{c<c'} w_{cc'}\, s_{cc'}^2(W;\mathcal{D})$ $(w_{cc'} > 0)$, which is the population counterpart of the ProtoNet metric loss under this Gaussian model. The encoders of the two protocols are, respectively, the maximiser of the source-domain objective $W_R = \arg\max_W J(W; \mathrm{src})$ and the maximiser of the pooled objective $W_H = \arg\max_W [(1-\lambda)J(W;\mathrm{src}) + \lambda J(W;T)]$, where $\lambda \in (0, 1]$ reflects the effective weight of the held-out domain T in the pooled training set.

Lemma 1 (pairwise correctness probability under population prototypes). Under the above setting and population prototypes, the probability that a class-c query is closer to the prototype of its own class (relative to c') is exactly

$$\Pr[\text{correct} \mid (c, c')] = \Phi\big(s_{cc'}(W;T)/2\big), \tag{1}$$

where Φ is the standard normal CDF. Projection onto the perpendicular bisector hyperplane of the decision boundary shows that this probability depends on W, Σ *only* through $s_{cc'}$ (the covariance components orthogonal to the normal have no effect), and that it is strictly increasing in $s_{cc'}$. (At finite K the sample prototypes are noisy and the discriminant normal is likewise random, so the exact pairwise probability is the orthant probability of two Gaussians $\Pr[B^\top D > 0]$, and is in general no longer a function of a single s; this noise is shared by the two protocols, and the formal experiments test the ordering at finite K empirically with $K \in \{1, 3, 5, 10, 20, 50\}$.)

Lemma 2 (the target-domain separability objective of the pooled encoder is no lower than that of the source-domain encoder). By the optimality of W_R, W_H,

$$\begin{aligned} J(W_H;T) - J(W_R;T) \;&\geq\; \tfrac{1-\lambda}{\lambda}\,\big(J(W_R;\text{src}) - J(W_H;\text{src})\big) \\ &\geq\; 0. \end{aligned} \tag{2}$$

This inequality bounds only the scalar objective $J(\cdot;T)$; it does not by itself bound any individual $s_{cc'}$ or the accuracy.

Theorem (evaluation-leakage bias is non-negative, two-class). Let A_H, A_R be the expected few-shot accuracies of the shared and of the fold-restricted protocol on T, and $\Delta := A_H - A_R$. Under the above assumptions, population prototypes and $C = 2$,

$$\Delta = \Phi\big(s(W_H;T)/2\big) - \Phi\big(s(W_R;T)/2\big) \;\geq\; 0, \tag{3}$$

with equality if and only if $s(W_H;T) = s(W_R;T)$. (For the conditional extension to the multi-class case $C > 2$ see Proposition 1 below; because changing W simultaneously changes the correlation structure of the several decision normals, dominance of the mean pairwise separability alone is not sufficient to guarantee dominance of the multi-class accuracy, so an explicit coupling condition is required.)

Corollary. (i) *No shift, no leakage*: if the source and the target domain are identically distributed and the maximiser is unique, then $J(\cdot;\text{src}) \equiv J(\cdot;T) = J_{\text{pool}}$, $W_H = W_R$ and $\Delta = 0$; evaluation leakage is a product of distribution shift. (ii) *Vanishing at zero shift, generally positive under shift*: Δ is continuous in the source/target parameters and takes the value zero on the no-shift set; when W_R is not a maximiser of the pooled objective (the general case in which the source-optimal and the target-optimal encoder differ) and (2) is strict, then $s(W_H;T) > s(W_R;T)$ and, in the two-class case, $\Delta > 0$. (iii) *Principled remedy*: to estimate cross-domain few-shot generalisation without bias, the training set for representation learning must be disjoint from the held-out evaluation domain, which is exactly the fold-restricted protocol. (At finite K the prototype noise lowers A_R, A_H together, and the K dependence of Δ is characterised empirically, with no closed-form monotonicity asserted.)

Proposition 1 (conditional non-negativity in the multi-class case $C > 2$). The two-class theorem above cannot be extended to the multi-class case by Lemma 2 alone, because changing W simultaneously changes the correlation structure of the several decision normals. We give a rigorous multi-class result under an **explicit sufficient condition**. For a true class c, define its standardised decision-margin vector relative to each competing class $c' \neq c$, $Y_c(W) \in \mathbb{R}^{C-1}$; under population prototypes $Y_c(W) \sim \mathcal{N}\big(m_c(W), R_c(W)\big)$, where the c'-th coordinate of the mean is $\frac{1}{2}s_{cc'}(W;T)$ and $R_c(W)$ is the standardised margin correlation matrix; class c is correct if and only if $Y_c(W)$ falls in the positive orthant. *If* (i) the same-class noise correlation structure is the same under the two protocols, $R_c(W_H) = R_c(W_R)$, and (ii) each pairwise separability is coordinatewise dominant, $s_{cc'}(W_H;T) \geq s_{cc'}(W_R;T)\ \forall c'$, *then* a coupling by identically distributed common noise on an auxiliary space yields class-by-class dominance of the positive-orthant probability, $P_c(W_H) \geq P_c(W_R)$, whence $A_H \geq A_R$, that is, $\Delta \geq 0$. The condition is sufficient but not necessary, and can be diagnosed with large samples from the target domain (by checking the change in $\widehat{R}_c$ against the signs of the increments of the several $\widehat{s}_{cc'}$); the multi-class empirical results in the main text do not depend on its holding, and it serves only as an explicit boundary of multi-class provability (the complete proof is given in Supplementary Information §S1).

Proposition 2 (an explicit shift-dependent lower bound on the leakage). We raise "the larger the shift, the larger the leakage" to an analytic lower bound. Define the **leakage-relevant shift** as the pooled-objective gap $D_{\text{pool},\lambda} := J_\lambda(W_H) - J_\lambda(W_R) \geq 0$ (with $J_\lambda = (1-\lambda)J(\cdot;\text{src}) + \lambda J(\cdot;T)$), which characterises "how much extra the separability objective can gain, relative to the source-domain-optimal encoder, once the target domain is let into representation training". Suppose the target-domain signal-to-noise ratio is bounded above on the feasible set by $S_T < \infty$ (guaranteed by the compactness and non-degeneracy of $\mathcal{W}$). Then, in the two-class case,

$$\Delta \;\geq\; \frac{\exp(-S_T^2/8)}{4\sqrt{2\pi}\, w\, S_T\, \lambda}\, D_{\text{pool},\lambda}, \tag{4}$$

where the right-hand side is zero when $D_{\text{pool},\lambda} = 0$ (no shift), is monotonically non-decreasing in the shift, and gives a strictly positive lower bound whenever the shift is strictly positive (proof in Supplementary Information §S1). **It must be emphasised that the MMD in the raw input space is not a sufficient measure of shift in Eq.** (4)**.** Counterexample: when the source and target class means are shifted in common by $a \neq 0$, the between-class margin and the nearest-prototype separability are unchanged, so $\Delta = 0$, yet the linear-kernel MMD$= a^2 > 0$. The raw MMD therefore enters the lower bound only under the additional calibration assumption that "it controls the leakage-relevant separability shift from below"; otherwise the relation of Δ to MMD should be reported as an empirical regularity rather than as an unconditional theorem—which is consistent with the observation in Section 3.10 that the MMD slope is not significant in the cross-dataset regression.

The falsifiable core given by this theory is $\Delta \geq 0$ (strict in the two-class case, and strict in the multi-class case under the condition of Proposition 1), together with the fact that the leakage is lower-bounded by the *leakage-relevant shift* rather than by the raw MMD. We test this on apple NIR, on the public OliveOil data, and on corn/tablet/mango/Indian Pines, **six datasets** in all (Sections 3.3, 3.9 and 3.10): the leakage is robustly positive across domains, whereas the raw MMD does not linearly predict its magnitude—in complete agreement with the theory and with its honest boundaries.

Empirical diagnosis of the conditions of the propositions and validation at finite K. In order to connect the theory with actual neural encoders, finite K and the multi-class regime, we diagnose the sufficient condition of Proposition 1 directly (script `28_g2_theory_diagnostic`). On the 13 leave-one-domain-out folds of the four multi-class datasets, estimating the pairwise separabilities $s_{cc'}$ of the several class pairs in the *target-domain embedding*, the proportion of cases in which condition (ii) of Proposition 1, "coordinate dominance" ($s_{cc'}(W_H;T) \geq s_{cc'}(W_R;T)$), holds **reaches a mean of** 0.862; particularly consistent is the fact that the fold with the lowest coordinate-dominance proportion (a certain high-shift quadrant of Indian Pines, only 0.40) **is precisely the only fold with negative leakage in Section 3.10**—that is, wherever the theoretical condition fails, the leakage there ceases to be positive, the two corresponding mechanistically. As for condition (i), the change in the standardised margin correlation structure is very small on the point-spectroscopy datasets (≤ 0.13) and larger on hyperspectral imaging, consistent with the stronger geometric rearrangement of that modality. In addition, we estimate $\mathbb{E}[\Delta]$ empirically in the linear-Gaussian model using **finite-K sample prototypes** (rather than population prototypes): throughout $K \in \{1, 3, 5, 10, 20\}$ we have $\mathbb{E}[\Delta] > 0$ (from $+0.020$ to $+0.006$), and the probability that a single episode has $\Delta \geq 0$ is 0.81–0.93, showing that non-negative leakage continues to hold robustly in the finite-K regime beyond the population-prototype limit.

Formal statistical tests and mechanistic validation on real embeddings. To raise the above descriptive diagnosis to formal inference, and to lift the finite-K argument from analytic simulation to *real neural embeddings*, we carry out two further tests (script `29_g2_theory_validation`). First, taking the coordinate-dominance proportion as a continuous proxy for the strength of the condition of Proposition 1, we test its fold-by-fold relation with the *actual leakage* Δ (K=5, obtained by resampling finite prototypes on real target-domain embeddings): the two are positively correlated (Spearman $\rho = +0.44$), in the direction predicted by the theory; the permutation test gives $p = 0.12$, not reaching significance, which we honestly attribute to the insufficient power caused by the limited number of folds (n=13), and to the fact that under real embeddings *all 13 folds have positive* Δ, so that the "failure fold" contrast is almost entirely absent. The coordinate-dominance proportion itself is robustly higher than the chance baseline 0.5: its mean is 0.813, and the 2000-resample percentile bootstrap 95% CI is [0.682, 0.931], whose lower bound is far above 0.5, providing formal interval evidence for condition (ii) of Proposition 1. Second, we resample finite-K prototypes on the *real target-domain embeddings* (the actual outputs of the shared encoder W_H and of the fold-restricted encoder W_R) and compute $\Delta(K)$ at each K: over $K \in \{1, 3, 5, 10, 20\}$, $\mathbb{E}[\Delta]$ is $+0.079, +0.079, +0.078, +0.073, +0.062$ respectively, with a per-fold positive proportion of 0.69–1.00. The non-negative leakage on real neural embeddings is **markedly larger in magnitude than in the linear-Gaussian simulation** (0.06–0.08 against 0.006–0.02), showing that the idealised model is a conservative lower bound and that the geometry of the real representation amplifies the leakage further. These two tests connect Propositions 1/2 from idealised statements to the actual multi-class, finite-K, neural-encoder experimental conditions of this paper.

6.4 ProtoNet-C: a source-centroid fusion prototypical network

The vanilla prototypical network (Snell et al., 2017) takes the mean embedding of the K support samples as the class prototype and then assigns a query sample to the nearest prototype under squared Euclidean distance. When K is very small (for example K=1), the single-sample prototype has high variance, and the rich class statistics available in the source domains go unused.

ProtoNet-C (C stands for centroid fusion) addresses this by linearly interpolating each episodic prototype $\mathbf{c}_k^{\text{task}}$ with the source-domain centroid $\mathbf{c}_k^{\text{src}}$:

$$\hat{\mathbf{c}}_k = (1 - \alpha)\, \mathbf{c}_k^{\text{task}} + \alpha\, \mathbf{c}_k^{\text{src}}, \tag{5}$$

where $\alpha \in [0, 1]$ is the fusion weight, $\mathbf{c}_k^{\text{src}}$ is the mean embedding of all source-domain samples of class k, and $\mathbf{c}_k^{\text{task}}$ is the K-sample mean of the support set of the current episode. At $\alpha = 0$ the model degenerates to vanilla ProtoNet; at $\alpha = 1$ it degenerates to a nearest-centroid classifier anchored entirely on the source prototypes.

A bias–variance perspective: the fusion formula (5) can be understood from the standpoint of the bias–variance trade-off. When K is very small, $\mathbf{c}_k^{\text{task}}$ is a high-variance estimate of the class centre; when the source–target domain gap is small, injecting $\mathbf{c}_k^{\text{src}}$ can reduce the estimation variance at the cost of an acceptable bias, and may thereby raise the expected classification accuracy (Ben-David et al., 2010). However, once the domain gap exceeds a certain threshold—as with cross-year distribution shift or a measurement-protocol mismatch—the bias introduced by the source centroid dominates, so that a fusion weight $\alpha > 0$ becomes harmful instead. In Section 3.6 we test this theoretical prediction with controlled experiments, in order to determine the appropriate value of α in the cross-year apple NIR scenario.

The encoder f_θ is a five-layer fully connected network (input 229, hidden dimensions 512-256-128-64, ReLU activation, dropout 0.3 (Srivastava, Hinton, Krizhevsky, Sutskever and Salakhutdinov, 2014)), projecting onto a 64-dimensional ℓ_2-normalised embedding space. Meta-training follows an episodic strategy: each episode samples a 3-way K-shot task from the source domains, and the Adam optimiser (Kingma and Ba, 2015, $lr = 10^{-3}$) is used to minimise the cross-entropy loss of the query predictions. Training runs for 500 epochs of 100 episodes each; the checkpoint with the highest macro F1 on a held-out source validation split is used for evaluation.

6.5 Baseline methods

Four baseline methods are compared in parallel with ProtoNet-C. The **SVM** baseline uses a C-support vector classifier (C-SVC, RBF kernel), with the regularisation parameter $C \in \{0.01, 0.1, 1, 10, 100\}$ and the kernel width $\gamma \in \{\text{scale}, 0.001, 0.01, 0.1, 1\}$ selected by grid search with 5-fold cross-validation on the source-domain validation set; at test time the classifier is refitted on the K target-domain support samples (with no re-selection of hyperparameters, the source-domain optimal hyperparameters being used). **PLS-DA** builds a PLS projection from the K support samples, with the number of latent components optimised by leave-one-out cross-validation within the range $[2, \min(10, K-1)]$; when $K \leq 2$ the number of latent components is fixed at 1. **Vanilla ProtoNet** is the special case of ProtoNet-C at $\alpha = 0$, using exactly the same fold-restricted encoder architecture and training pipeline as ProtoNet-C, and is reported for all $K \in \{1, 3, 5, 10, 20, 50\}$ (a single random face position is uniformly used for evaluation on every fold, consistent with the single-face acquisition protocol of the 2018/2019 folds). **Nearest centroid** classifies each query by the nearest support-set mean in the embedding space, with no meta-training at all, and serves as the simplest prototype-based reference.

It should be noted that this paper does not include traditional unsupervised domain-adaptation methods such as transfer component analysis (TCA) (Pan and Yang, 2010) and subspace alignment (SA) (Fernando et al., 2013) among the baselines. Such methods generally need access to the full unlabelled target-domain data when performing distribution alignment, which stands in fundamental protocol-level asymmetry with the episodic K-shot evaluation framework of this paper—which permits access only to K labelled target-domain support samples at inference time—so that a direct comparison would introduce an informational advantage favourable to the traditional methods. Including TCA/SA under the present evaluation protocol therefore cannot achieve a fair comparison; incorporating them into a feasible future comparison framework is an extension worth considering, as discussed in Section 4.

6.6 Alpha ablation: the source-prototype fusion weight

To determine whether source-centroid fusion is beneficial on each fold and to identify the optimal α, three representative configurations are compared: **configuration A** uses $\alpha = 0.4$ uniformly on all three LOEO folds; **configuration B** uses $\alpha = 0.4$ on the 2018 and 2019 folds and $\alpha = 0$ on the 2025 fold (the motivation being that source-domain centroids trained on single-face data may fail to represent the multi-face 2025 target distribution); **configuration C** sets $\alpha = 0$ everywhere, which is equivalent to vanilla ProtoNet with a fold-restricted encoder. This ablation directly tests whether the injection of source centroids is beneficial or harmful under cross-year distribution shift.

Here $\alpha = 0.4$ serves as a representative non-zero fusion weight, determined by a coarse-grained grid search over $\alpha \in \{0.1, 0.2, 0.4, 0.6, 0.8\}$ on the source-domain validation split: on the source-domain validation sets of the LOEO-2018 and LOEO-2019 folds, $\alpha = 0.4$ gave second-best to intermediate validation performance within that candidate set, while also agreeing with the convention in the domain-adaptation literature of setting the fusion weight mostly in the interval $[0.3, 0.5]$ (Pan and Yang, 2010). The complete continuous sensitivity curve over $\alpha \in [0, 1]$ is given in Section 3.6 (Fig. 4).

6.7 Face-mean ablation

The 2025 subset provides about five measurement positions per apple, enabling us to test systematically how multi-face aggregation affects classification. We define a 2×2 design crossing the support-side strategy (single face versus multi-face mean) with the query-side strategy (single face versus multi-face mean): $S_{\text{single}}Q_{\text{single}}$ (baseline), $S_{\text{multi}}Q_{\text{single}}$ (support-side mean only), $S_{\text{single}}Q_{\text{multi}}$ (query-side mean only) and $S_{\text{multi}}Q_{\text{multi}}$ (means on both sides, the principal method). This 2×2 design allows a main-effect decomposition of the mean gain and an assessment of the interaction. The ablation is carried out on the LOEO-2025 fold at $K = 5$ with 50 repeated episodes, because 2025 is the only year with multi-face data. An SVM-Kshot control is also included, in which each face of a multi-face apple serves as an independent training sample, giving the SVM access to about $5K$ effective training instances.

6.8 Design of the multi-seed variance analysis

The precondition for comparing a NIR benchmark across papers is that the individual sources of evaluation uncertainty be explicitly disentangled and their magnitudes given separately, which also echoes the calls in the machine-learning community for reproducibility reporting standards. We divide the sources of randomness into three layers: (i) the **data-splitting layer**—LOEO-Year is a deterministic hold-out at the granularity of years, and there is no degree of freedom of random splitting; (ii) the **encoder-training layer**—for each LOEO fold the ProtoEncoder is retrained from scratch independently with 5 seeds (42–46), the remaining training hyperparameters (architecture, Adam $lr = 10^{-3}$, CosineAnnealing, patience=12×100 episodes, n_{way}=3, k_{support}=5, q_{query}=10, temperature=10) being exactly as in Section 6.4; (iii) the **episode-sampling layer**—each encoder then draws support/query episodes with 50 independent seeds (42–91).

This constitutes **3 folds × 5 encoder seeds = 15 independent encoders**, each encoder then being expanded into a full-factorial design of **6 K values × 50 episode seeds × 4 methods (vanilla ProtoNet $\alpha = 0$, SVM, PLS-DA, nearest centroid) = 18,000 evaluations**; within it, the ProtoNet evaluation mode is strictly consistent with the "unified single-face query protocol" of Section 6.5. The first 16 characters of the SHA-256 hash of every encoder checkpoint (.pt), the source-domain validation accuracy and the training-termination episode are all recorded. The complete experimental record (encoder metadata, long-format episode-level F1 records, the summary variance-decomposition table and the direction-consistency table) is released with the supplementary material of the paper.

The evaluation variance is quantified along two axes: (a) the **encoder-level standard deviation** (between-encoder SD)—for each encoder the macro F1 is first averaged over the 50 episodes, and the standard deviation across the 5 encoders is then taken; (b) the **episode-level standard deviation** (within-encoder SD)—the standard deviation of the 50 episodes within each encoder, then averaged over the 5 encoders. The ratio of the two reflects the relative magnitude of the perturbation of the results caused by "retraining the encoder with another seed" as against "drawing another set of random episodes", and thereby delimits the regime in which the "50-repetition std" is valid as a single uncertainty index (see Section 3.2 for details).

6.9 Hardware, deterministic configuration and the two-stage experiments

The experiments of this paper were executed in two stages, which are disclosed separately. The **Formal stage (all core results reported in the main text)** was run under CUDA on a single NVIDIA RTX 4090 (24 GB), covering the evaluation-leakage quantification, the K-shot method comparison, the α ablation, the face-mean ablation, the statistical tests, the per-class F1, the external validation, the negative controls, the uncertainty calibration, the interpretability analysis and the composite LLH-GBID evaluation; it uniformly used the 5 fixed model seeds $\{20060515, 20041210, 19810915, 2023, 2024\}$ and enabled `torch.use_deterministic_algorithms(True)`, `cudnn.deterministic=True` and `cudnn.benchmark=False`; the input data were locked to a frozen canonical version by SHA-256 checksums. The **Pilot/supplementary stage** was run on Apple Silicon MPS and served only for feasibility verification and for the two-level encoder–episode variance decomposition (Section 3.2, seeds 42–46 / 42–91); the MPS backend does not provide deterministic kernels for a few individual operators, so the results of that part are **explicitly labelled as coming from MPS hardware**, serve only as a methodological supplement, and do not carry any core conclusion. All reported standard deviations reflect only model stochasticity (encoder initialisation and episode sampling), and **do not include hardware-layer numerical jitter**; any population-level statistical inference across independent harvest seasons requires validation on future independent datasets. For every formal run, the GPU model, wall-clock time, GPU-hours, peak GPU memory, energy consumption and carbon-dioxide equivalent were recorded by `track_compute` and collated into the `compute` sheet of each script's xlsx file.

6.10 Statistical tests

The principal method comparison (ProtoNet-C versus SVM) takes the **fold** as the primary unit of analysis ($n_{\text{primary}} = K_{\text{outer}} = 3$): within each fold the macro F1 is first averaged over the 50 repeated episodes, a paired test is then performed across the 3 folds, and the results are aggregated across the 5 model seeds for reporting. A paired Student's t test and a Wilcoxon signed-rank test (Wilcoxon, 1945) are applied together to assess significance, and Cohen's d (Cohen, 1988) quantifies the effect size (convention: $|d| < 0.2$ negligible, 0.2–0.5 small, 0.5–0.8 medium, ≥ 0.8 large). For comparisons spanning multiple K values, the Benjamini-Hochberg procedure (Benjamini and Hochberg, 1995) controls the false discovery rate at 5%. In addition, the between-fold analysis (LOEO-2018 and LOEO-2019 versus LOEO-2025) provides a further independent check on the direction of the conclusions, because the two single-face folds and the 2025 fold do not overlap in their data sources.

Author contributions (CRediT)

Panlin Li: Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Visualization, Writing—original draft. **Qingli Zhang**: Investigation (sample collection and spectral measurement), Data curation, Validation. **Qitai Jiao**: Resources (near-infrared acquisition instrument and measurement platform), Investigation (spectral acquisition). **Jiamei Sun**: Investigation (sample collection), Data curation, Visualization. **Hua Huang**: Conceptualization, Supervision, Project administration, Funding acquisition, Writing—review & editing.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

All core method implementations of this paper (evaluation-leakage quantification, the four-encoder decomposition of $\Delta_{\mathrm{sup}}/\Delta_{\mathrm{unsup}}$, ranking flips, the cross-domain mixed-effects statistics, the theoretical diagnostics and the de-risking controls), the preprocessing pipeline, the dataset card and the reproduction guide have been released in a public GitHub repository: `https://github.com/2004lryan/016representation-layer-eval-leakage` (MIT licence). The datasets used for the cross-domain validation and for the supervised/unsupervised leakage decomposition are all public standard benchmarks: OliveOil spectra, corn NIR (Cargill corn(Eigenvector Research, 2000)), pharmaceutical tablet NIR (the 2002 IDRC shootout(IDRC, 2002)), mango NIR(Anderson et al., 2020) and Indian Pines hyperspectral(Baumgardner et al., 2015), all of which can be obtained through standard channels (the repository file `data/DATA.md` records the source, licence and SHA-256 checksum of each set). The nine-domain apple near-infrared spectral dataset that supports the conclusions of this paper is in-house data of Xinjiang Agricultural University under restricted access (managed access), and is **not yet publicly released**—public archiving of it has not so far been authorised by the data owner; researchers who wish to reproduce or extend the experiments of this paper may apply to the corresponding author on a case-by-case basis, in accordance with the data-sharing policy of Xinjiang Agricultural University. The dataset card for this apple dataset (containing per-domain descriptive statistics, the splitting logic by fruit ID and the SHA-256 checksums) has been released together with the public code repository above. In the journal's single-blind review process, an anonymised, review-only data access link can be provided to the reviewers through the editorial office in order to support independent verification.

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